PROGRESS BRIEFING · OCTOBER 2026

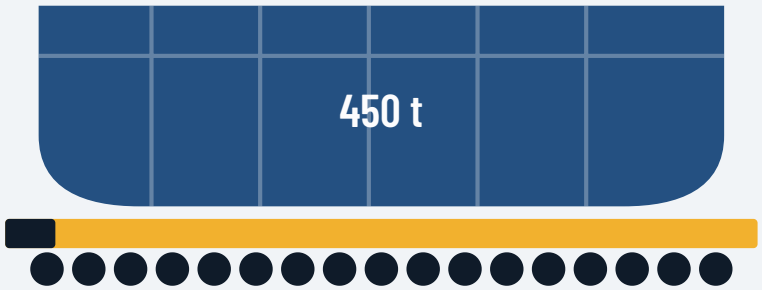
One big transporter for a heavy hull block, or several small ones coupled?

Single-carrier capacity or cooperative transport? Shipyard transporter fleets are sized under one service specification. They are priced from tender awards.

Service-constrained fleet sizing for single-carrier and cooperative block transport in shipyards

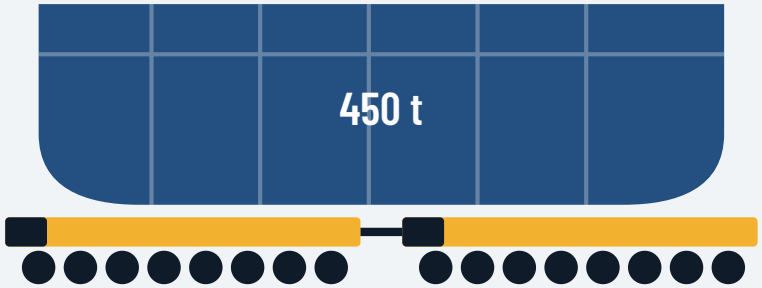
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Harbin Engineering University · Nanyang Technological University
Target journal: Ocean Engineering (manuscript updated to version R03, being finalised)

A · 1 × 500 t



1 vehicle · 4 crew

B · 2 × 270 t



2 vehicles · 8 crew, wait for both, then couple

The whole study in one minute

1

The question

Coupled transporters can move blocks heavier than any single vehicle. So what should a yard buy? Heavy units, light units that couple, or a mix? Existing studies schedule a given fleet.

2

What we did

Nine tiers and light–heavy mixes face one service standard. It needs ≥ 95% on time and no block over 120 min late. Each fleet gets the smallest qualifying count found. We use 36 conditions of 30 days. Prices: 33 tender awards and official wages.

3

The answer

Under shift labour there are 1,620 decisions. In **91.0%**, the least-cost fleet couples at most one block in ten. Where some tier carries every block, all 1,350 decisions do. In 13 further conditions, light units with matched heavy units win 95%.

4

Why

Coupling holds two or three vehicles and crews for the whole trip. Repeated service and alignment are a median 90% of a light fleet's extra transporter-time. Partner waiting is 2%. Under shift labour, a light tier can couple only 2–18% of its workload.

> **Coupling is a remedy for exceptional blocks, not a purchasing strategy.**

Sources: workload Roh & Cha (2011). Block masses Jiang et al. (2021), Roh & Cha (2011), Liu et al. (2022). Handling Liu et al. (2022), Kim & Joo (2012). Prices: 33 public award notices (2022–2026, three procurement platforms). Wages: National Bureau of Statistics of China (2026), Table 7; China Statistical Yearbook 2025, table 4-12. Boxes 3 and 4 are results of this study (paper abstract, §5).

The order of this briefing (click to jump)

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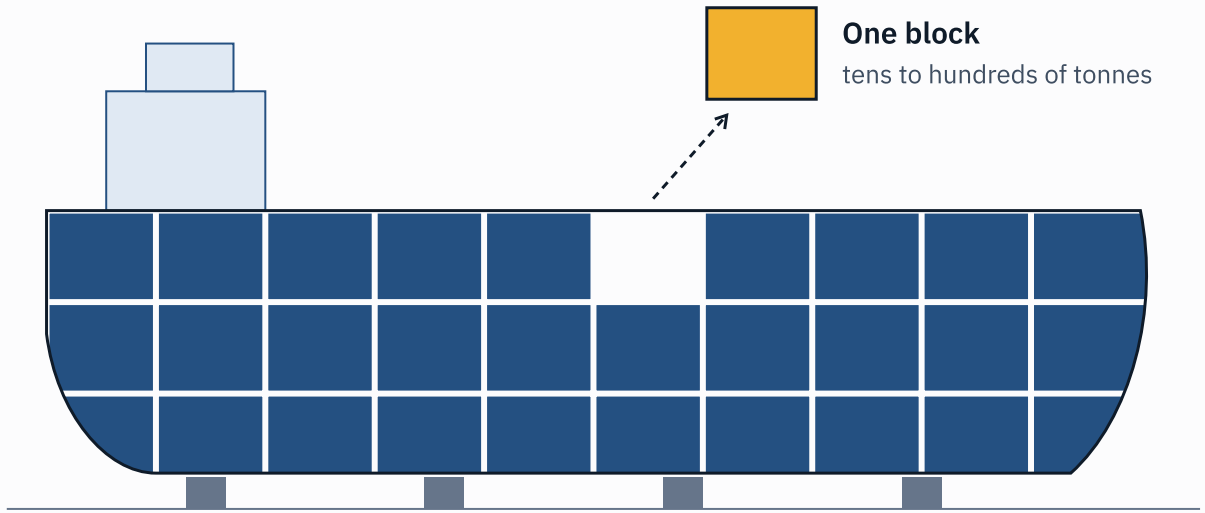
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Ships are built in blocks, then joined in the dock



A hull in the dock, welded together from many blocks (schematic)

- A large ship is divided into many **blocks**. They are built, outfitted and painted **in parallel** in different shops. Then they are joined in the dock.
- A typical block is about 15 m × 15 m × 5 m. It weighs 100–300 t, and some exceed 500 t.
- The price: blocks are **moved many times**. A late move holds up the next process.

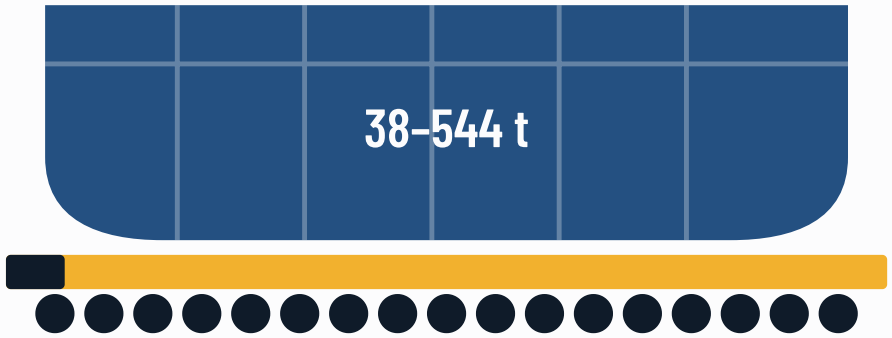
This study is about those moves. Which transporters, and how many, move them at least cost?

Sources: block size and mass Park & Seo (2012), p. 390.

Each block is moved several times between shops, yards and the dock

A typical route of one block

Assembly shop → Outfitting shop → Paint shop → Block yard → Assembly yard → Dock

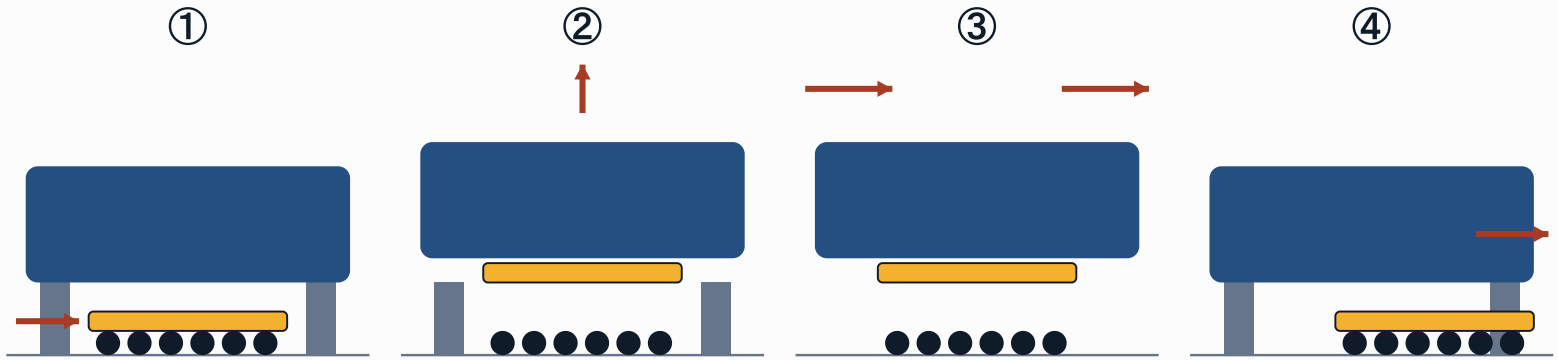


Self-propelled hydraulic transporter, rated 200–550 t

- **Block masses** in published samples: 38–450 t, 204–399 t, 260–544 t
- **Transporters**: self-propelled hydraulic units. Fleets in the literature are rated 200–550 t
- **Price**: about CNY 1.79–3.92 million per unit for 200–550 t. Ex VAT, fitted to tender awards
- **Crew**: four or five workers per transporter in practice, a driver and guiders. This study uses the lower end. Each transporter has its own crew of four: one driver, three guiders
- **Workload**: on one recorded day, four transporters moved 97 blocks in 16 h

Sources: masses Jiang et al. (2021) Tables 5, A1, A2. Liu et al. (2022) S1 data. Roh & Cha (2011) Table 2. Fleet ratings Jiang et al. (2021) Table 5. Roh & Cha (2011) Table 1. Liu et al. (2022) Table 4. Crew Park & Seo (2012), p. 392. Workload Roh & Cha (2011), p. 3246. Prices 33 public tender awards (2022–2026).

How a transporter moves a block: drive under, lift, drive, set down



① Drive in empty

The block rests on supports. The vehicle drives into the gap underneath

② Lift hydraulically

The platform rises and lifts the block off the supports. This is loading

③ Drive loaded, slowly

It carries the block to the destination. It drives at about half its empty speed

④ Set down, leave

The platform lowers the block onto new supports: unloading. The empty vehicle heads for the next block

Rated capacity

Sold by tier. One unit cannot carry a block above its rating

Speed

270 t model: 12 km/h empty, 6 km/h loaded

Loading / unloading

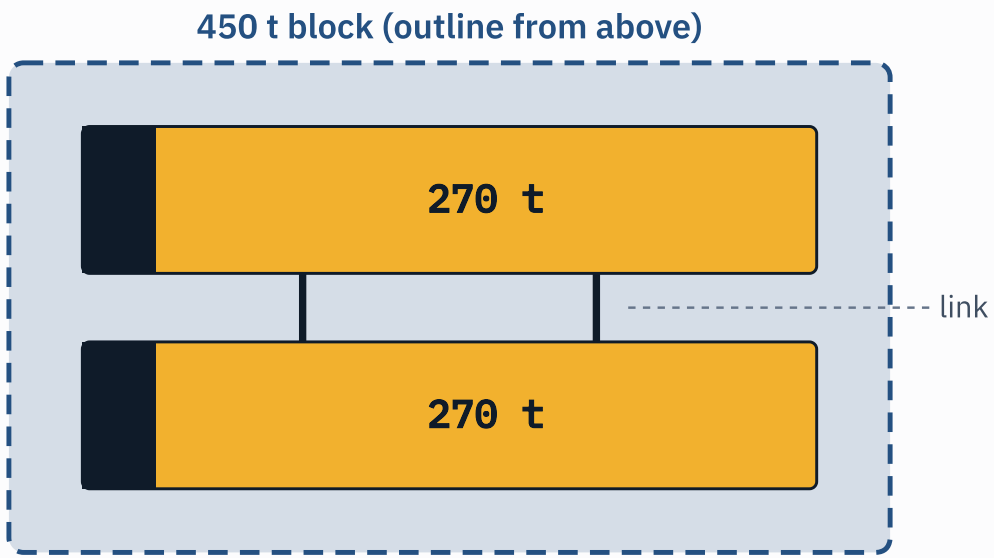
10 min each. A mass-dependent formula also exists. Loading in practice often takes 30–60 min

Crew

Four or five per vehicle in practice: a driver and guiders. Here each has its own crew of four: one driver, three guiders

Sources: speeds from the Suzhou Dafang DCY270 data sheet (accessed 2026-09-24). 10 min each: Liu et al. (2022) §4.1.1. Mass-dependent formula: Kim & Joo (2012) §4. 30–60 min: Liu et al. (2022) §4.2.2. Crew: Park & Seo (2012), p. 392.

Coupling: two or three units side by side drive as one vehicle



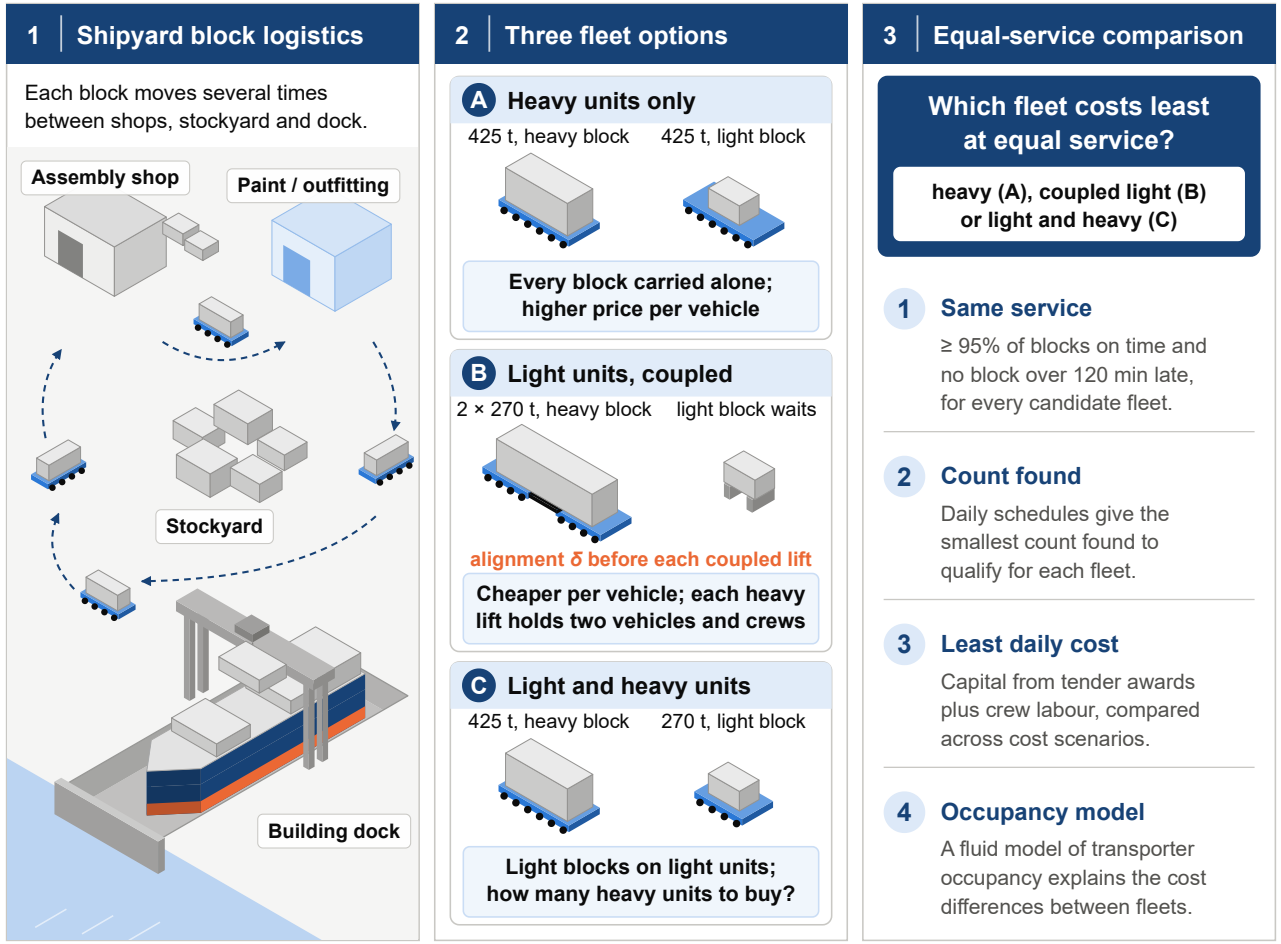
270 + 270 = 540 t ≥ 450 t

- **Capacities add**: two 270 t units lift up to 540 t, three up to 810 t. The team runs under synchronised control
- **All must be there**: the first to arrive waits for the last
- **Alignment takes time**: no measured value is published. We assume 10 min and test 0–40 min
- **Held throughout**: from loading to unloading, each unit and its crew can do nothing else
- At most three units per team (assumed). Teams of at most two were also tested

Coupling lets light units carry heavy blocks. But one block then takes two or three times the vehicles and crews. It saves capital and costs vehicle-time.

Sources: synchronised control, TII Group self-propelled transporter brochure (0317-EN), p. 21. Big tasks by several flatcars, Jiang et al. (2021), constraint (24). Coupled-use orders on the Suzhou Dafang website. Alignment time and team size are our assumptions.

The problem in one figure: block logistics, three fleet options, one comparison



- ① **Block logistics**: each block moves several times between shops, stockyard and building dock
- ② **Three fleet options**: A has heavy units only. B has light units coupled for heavy blocks, with alignment δ before each coupled lift. C has light units with a few heavy units. The question is how many
- ③ **The comparison**: every fleet gets the same service. Scheduling finds its smallest qualifying count. Then daily costs are compared. A fluid occupancy model explains the differences

> **Every fleet is held to the same service before costs are compared.**

Sources: paper Fig. 1 (schematic). The yard is generic, not the layout of any real yard.

Earlier work schedules coupling and counts vehicles, but never weighs coupling against larger carriers

Shipyard block logistics

- Less empty travel and interference Roh & Cha 2011
- Heterogeneous transporters, delivery limits Kim & Joo 2012; Joo & Kim 2014
- Workload balance, set-ups, precedence, routing Park & Seo 2012
- Paths: distance and number of turns Wang et al. 2020
- Energy and time windows, barge loading, stockyard sequencing Guo et al. 2024; Kweon et al. 2026; Wang & Yin 2026
- More units above 320 t, from fleet utilisation Liu et al. 2022

Cooperative transport

- Vehicles synchronised for one large cargo Hu & Wei 2018
- Multi-vehicle one-cargo for 200–500 t blocks, time and CO_2 Jiang et al. 2021
- Shipyard transporters combined to cover a block, selected daily from a given set Shin & Bak 2014
- One job-shop transport needing several robots Gayon et al. 2025
- Operation, movement and load synchronisation Drexl 2012; Soares et al. 2024

Vehicle numbers and fleet mix

- Shipyard vehicle counts compared. One type, divisible tasks Zhou et al. 2025
- Types and numbers chosen with routes, with fixed costs Golden et al. 1984; Hoff et al. 2010; Koç et al. 2016
- AGV and mobile-robot fleets Vis et al. 2005; Vis 2006; Fragapane et al. 2021; Aziez et al. 2022
- Synchronised visits under uncertainty Hashemi Doulabi et al. 2020
- Robot-team composition from simulated schedules Pan & Yu 2024

Analogues, service and search

- Couplable warehouse robots: identical units, only how many Albert et al. 2025; Chaikovskaia et al. 2026
- Modular transit coupling along a corridor Cao et al. 2022
- Fleet size at a stated service level Eskandari & Mahmoodi 2016
- Production scheduling with conflict-free AGV routes Nishi et al. 2011
- ALNS operator performance depends on the problem Voigt 2025

The gap: shipyard work schedules coupled loads and compares vehicle counts. Routing and robot-team work link resource choice to performance. One comparison is still open. For indivisible blocks at the same service, should a yard buy single-carrier tonnage or couple? We keep teams, alignment, empty travel and due dates in daily scheduling. Fleets are priced from tenders and crew labour. We test how far decisions rest on search and sampled days.

Sources: full references on the last slides (paper §2, Table 2).

Three research questions, and what makes them hard

(i) What to buy §5.1, §5.3

Which fleet sizes and capacity mixes minimise capital and labour costs under one service specification?

(ii) When coupling can compete §3.3, §5.2

Under what workload and cost conditions can cooperative transport compete with single-carrier capacity?

(iii) How service is specified §5.4

How does a lateness cap affect fleet size and composition?

Checks of the findings §5.5

How far do search error, sampled days, parameters and the candidate set move the findings?

Four complications

- **Service**: a fleet can meet an on-time target by letting its hardest blocks slip. So the answer depends on how service is specified
- **Mixes**: light units with a few heavy units offer a middle way between the extremes
- **Insurance**: coupling is often argued for as cover for a broken heavy unit
- **Search error**: counts come from a heuristic. A rival that qualifies with one transporter fewer can overturn a cost ranking

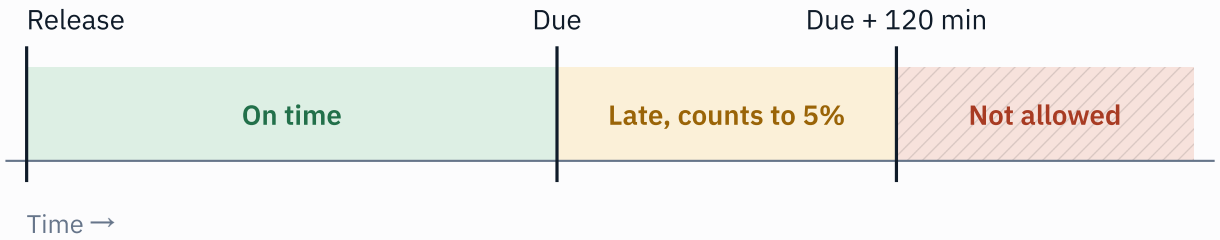
Sources: paper §1 (research questions (i)–(iii)).

Core idea: equal service first, then compare costs

≥ 95% on time · T_{max} = 120 min

An analogy

Two bus fleets cannot be compared on price alone. Ten buses run 80% on time, and twenty minibuses run 99%. First give both the same punctuality. Then see which is cheaper.



- Scheduling finds each fleet's **smallest qualifying count, K***. K* comes from a heuristic, so it is an upper bound on the true minimum.
- Then K* units are costed: capital plus crew labour per day.

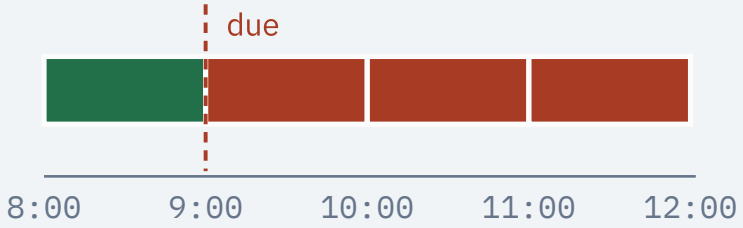
- **95% on time**: at least 2,765 of the 2,910 blocks of 30 days. At most 145 are late.
- **T_{max} = 120 min**: no block may finish more than 120 min late. That is one release-batch interval. So a late block cannot overrun the next batch.

Sources: the 95% target and T_{max} are our choices. The literature has no common service standard. T_{max} is tested from 30 min to no cap. The target is tested at 90% and 98%. Paper §3.1, §5.4.

Why not divide total work by one vehicle's hours?

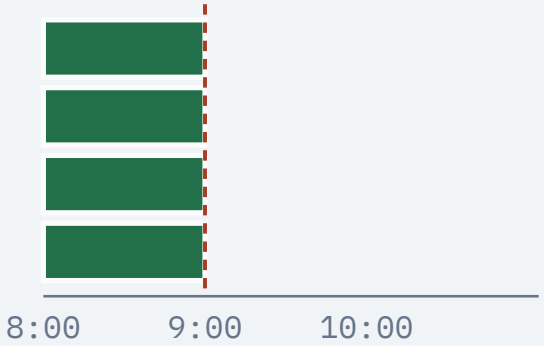
Example: four blocks ready at 8:00, all due by 9:00, each move 1 h

Totals only: 4 vehicle-h ÷ 16 h = 0.25 → one vehicle?



3 blocks late

By due time: four vehicles at once



All on time, but it takes 4 vehicles

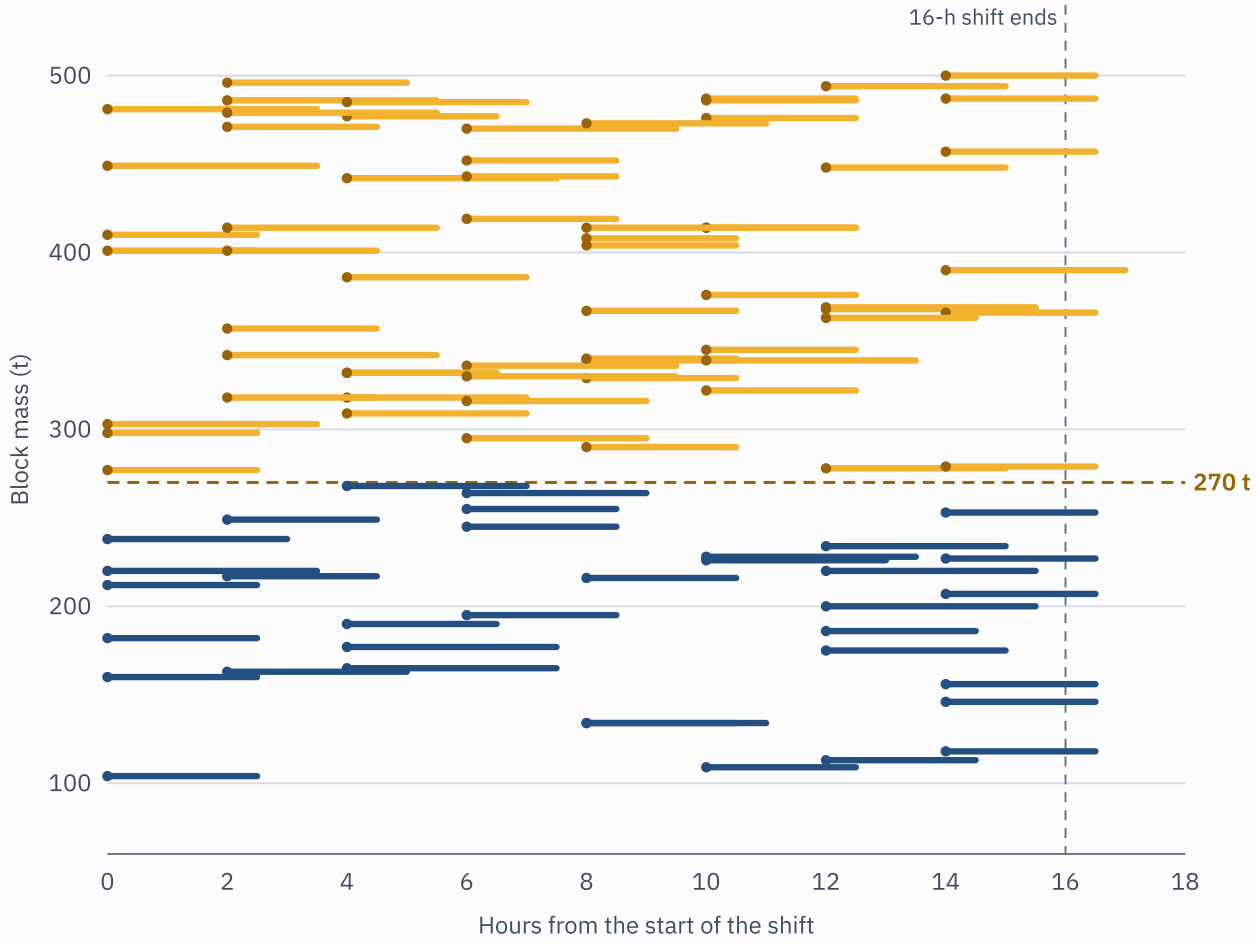
- **Work arrives in batches** with their own due times, so demand is uneven
- **Vehicles drive empty between tasks**, and the order changes that distance
- **A coupled lift needs several free vehicles at once**. One late arrival holds the team.

So each day's **actual timetable** must be built. The workload-only count is K_p. It is within one transporter of the scheduled count for 319 of 384 sized fleets. Under tight due dates, fleets that couple over 60% of blocks need more. On average they need 10.2% of K* more.

Sources: the example is illustrative. The figures 319 of 384 and 10.2% are results of this study (paper §5.2, Fig. 8c).

One working day: 97 blocks in eight batches, each with a due time

One generated day (uniform masses · short handling · empirical due dates)



- **97 blocks in 16 h**: the size of a recorded working day (8:00 to midnight)
- **Each line is a block**. The left end is release, the right end the due time. Height is mass.
- **Due times**: 150, 180 or 210 min after release, with probabilities 0.6 / 0.2 / 0.2
- **Release**: a batch every 2 h, about 12 blocks (assumed)
- **Orange** blocks exceed 270 t. A 270 t fleet must couple all of them (62 that day).

Each condition has 30 such days. Every fleet faces exactly the same 30 days.

Sources: 97 blocks and 8:00–24:00 from Roh & Cha (2011), p. 3246. Due times from Roh & Cha (2011) Table 2. Of 10 blocks, 6, 2 and 2 had 150, 180 and 210 min. Batch release is our assumption.

Who may carry together? Only minimal feasible teams

The team's ratings must add up to at least the block's mass. Removing any member must make it fall short. Extra units would only waste vehicles and crews.

$$\Sigma\, Q_k \geq m_i \quad \cdot \quad \Sigma\, Q_k - \min Q_k < m_i \quad \cdot \quad |S| \leq 3$$

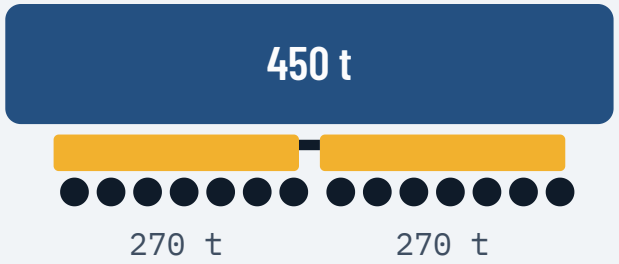
- A block one unit could carry may still go to a light team. This is flexible coupling, the main rule. The overweight-only rule is compared separately.
- For a homogeneous fleet the two rules coincide. Only light-plus-heavy mixes differ.
- Every minimal team is assumed physically feasible. Control, load sharing and road width are not modelled. This favours coupling.

Try it

Transporter tier (rated capacity)

200 t | 250 t | **270 t** | 300 t | 325 t | 380 t | 425 t | 500 t | 550 t

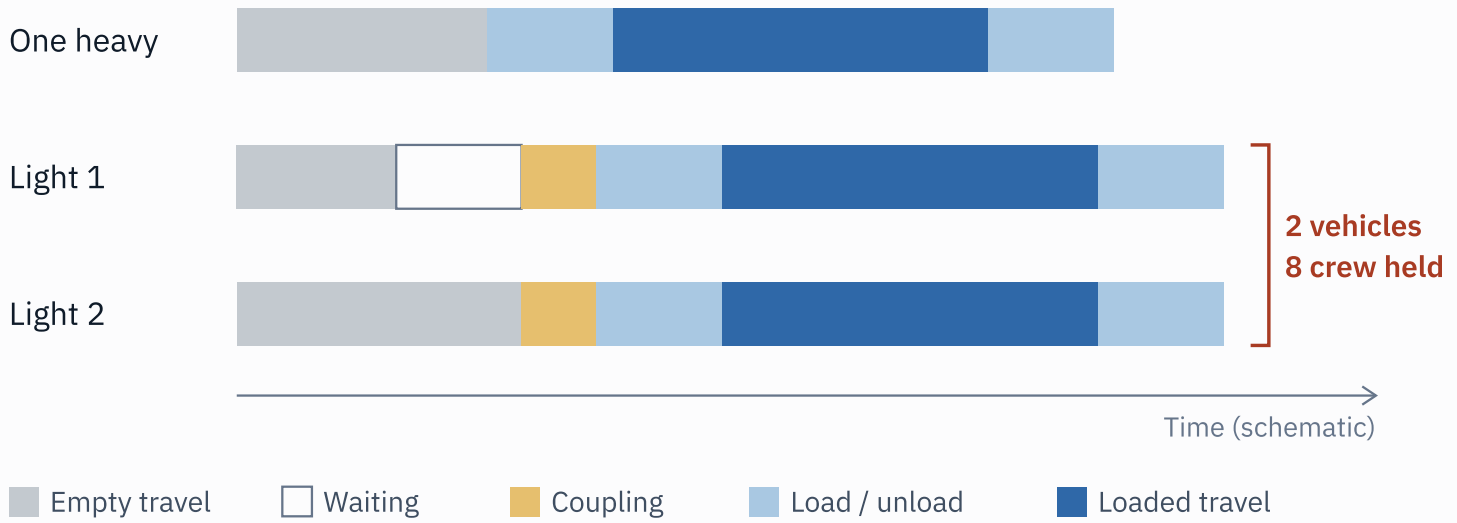
Block mass — 450 t



2 coupled: 2 × 270 = 540 t ≥ 450 t. One fewer gives only 270 t < 450 t. Holds 2 vehicles · 8 crew. Coupling adds 10 min.

Sources: big tasks by several flatcars, Jiang et al. (2021), constraint (24). The overweight-only rule follows Jiang et al. (2021). At most three units is our assumption. Two were also tested.

The price of coupling: all members meet and are held throughout

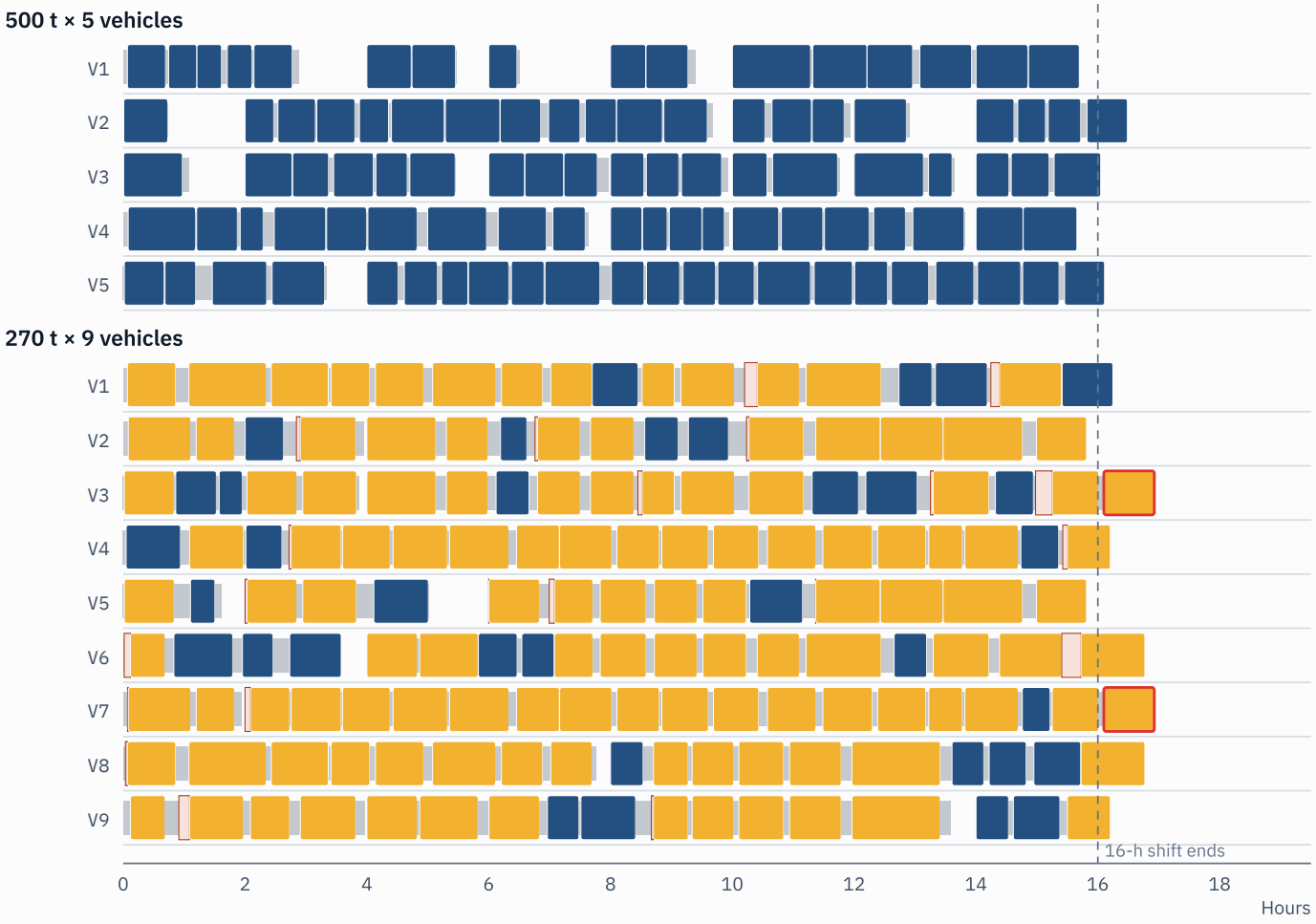


One move up close

- **Alone**: the vehicle loads on arrival. It leaves right after unloading.
- Example: a 420 t block takes 43 transporter-minutes on one 425 t unit. Two coupled 270 t units take 110. The extra is the second trip (43), alignment (2 × 10) and waiting (4).
- **Coupled**: the first unit waits for the second (white). Then both align (yellow).
- A light block waits for one of the units. It finishes at 98 min instead of 43.

Sources: illustrative times, as in Fig. 3 of the paper. Alone: 43 min = 10 empty + 33 service. The 10-min alignment is our assumption.

Same 97 blocks: two fleets' timetables



Empty travel · Carried alone · Coupled (incl. coupling) · Waiting for team · Late

	500 t × 5	270 t × 9
Vehicles K*	5	9
On time that day	97 / 97	96 / 97
Blocks coupled	0	62
Vehicle work (vehicle-h)	70.0	145.0
· empty travel	5.7	16.7
· waiting for team	0.0	1.9
Worst delay (min)	0	26
Fleet price P (270 t = 1)	8.15	9.00
Daily cost (labour-h, θ = 12.4)	421	688

The 270 t fleet needs nine units and twice the vehicle-hours. It couples 62 blocks. Here even its capital is higher.

Rows are vehicles, bars are tasks. Hover over a bar for the block. Both are stored schedules at each fleet's smallest qualifying count, re-simulated.

Sources: schedules of this study (uniform masses · short handling · empirical due dates, first day).

Far too many timetables to try, so we search cleverly

$$97! \approx 10^{152}$$

The order of 97 blocks alone has about 10^{152} possibilities. Teams are not yet counted.

So we use **adaptive large neighbourhood search (ALNS)**. It improves a workable timetable step by step.

1 Destroy

Remove some blocks. There are 5 ways: random, worst, related, a team or a route. Two more switch single ↔ coupled.

→

2 Repair

Reinsert them (greedy or regret-2). Choose a vehicle or team for each.

→

3 Accept

Keep the new timetable if it is better. Now and then accept a worse one, so the search does not get stuck.

→

4 Learn

Record which destroy and repair moves pay off. Use them more often. That is the "adaptive" part.

↻ 786 rounds per run, 38 s on average. Each run repeats exactly from its seed. The start is the best of 5,000 random chronological constructions.

How two timetables are ranked

① blocks over 120 min late → ② late blocks → ③ total lateness → ④ crew time. One more block on an earlier item outweighs any later gain. A team's members are updated together at every step. So coupled lifts stay synchronised.

Sources: ALNS framework and adaptive weights, Ropke & Pisinger (2006) and Pisinger & Ropke (2007). Operator types, Voigt (2025). Team dissolution, the two mode operators and team choice on insertion are specific to coupled lifts. Iteration and construction counts are our settings. Search depth is checked with counts after constructions only, 100 and 300 rounds. Paper §3.4, Supplementary S1.

Capital and labour on one scale: labour-hours

Vehicle

CNY 2.21 M

Price of one 270 t transporter (ex VAT, fitted to award prices)

→

CNY 329 k / yr

Spread over 8 years at 4% interest (capital recovery factor 0.1485)

→

1,316 CNY / day

250 working days a year

Labour

79,182 CNY / yr

Average wage of manufacturing production workers in China (NBS, 2025)

→

× 1.3 ÷ 2,000 h

Plus 30% employer on-costs (social insurance etc.), 2,000 hours a year

→

51.5 CNY / labour-h

Cost of one worker for one hour

Price ratio θ

θ = 1,316 ÷ 51.5 = 25.6

One 270 t transporter's daily capital cost = 25.6 labour-hours

Compare

64

Each transporter's own crew: a driver and three guiders. Per day: 4 × 16 h = 64 labour-hours.

Total cost

C = θ·P + L

P: the fleet's price in 270 t units. L: the fleet's labour-hours per day.

With Shanghai manufacturing wages (CNY 120.1 per labour-hour), θ = 11.0. Over all combinations of life, interest, working days, on-cost and wage, θ lies in 5.1–30. Every result is reported over that range.

Sources: 270 t price fitted to 33 public tender awards (ex VAT). National wage: National Bureau of Statistics of China, 2025 average wages, Table 7, released 15 May 2026. It covers production workers in large manufacturing enterprises: CNY 79,182. Shanghai wage: China Statistical Yearbook 2025, Table 4-12, manufacturing: CNY 184,833. On-cost follows official contribution rates, ≤ 39.4% in total. Rates: Shanghai HRSS 2025, State Council 2019 No. 13, MOHRSS 2015 No. 71, MOHURD 2016 No. 74. Crew of four: the lower end of four or five per transporter, Park & Seo (2012), p. 392. Service life, interest, working days and 2,000 h a year are our settings. θ is also tested over 0.5–5,000 (paper §3.2, Table 3).

Four ways to count labour, 6,480 decisions in all

Shift labour

64 · K

Every transporter has a crew of four for the 16-h day. This is the primary measure. It is the staffing a yard must provide, and it grows with K.

Shift + overtime

64·K + after-shift

Adds 1.5 × 4 crew-hours per transporter-hour after the shift.

Crew-vehicle hours

4 × vehicle-hours

Four crew for the hours each transporter actually works. Work includes empty travel, waiting and service.

Crew-team hours

4 per team when coupled

Empty travel and waiting count per vehicle. A coupled lift counts only four per team. It is as if one crew drove the coupled team. This favours coupling most.

5 price curves × 9 values of θ × 4 labour measures = 180 cost scenarios per condition · × 36 = 6,480 decisions

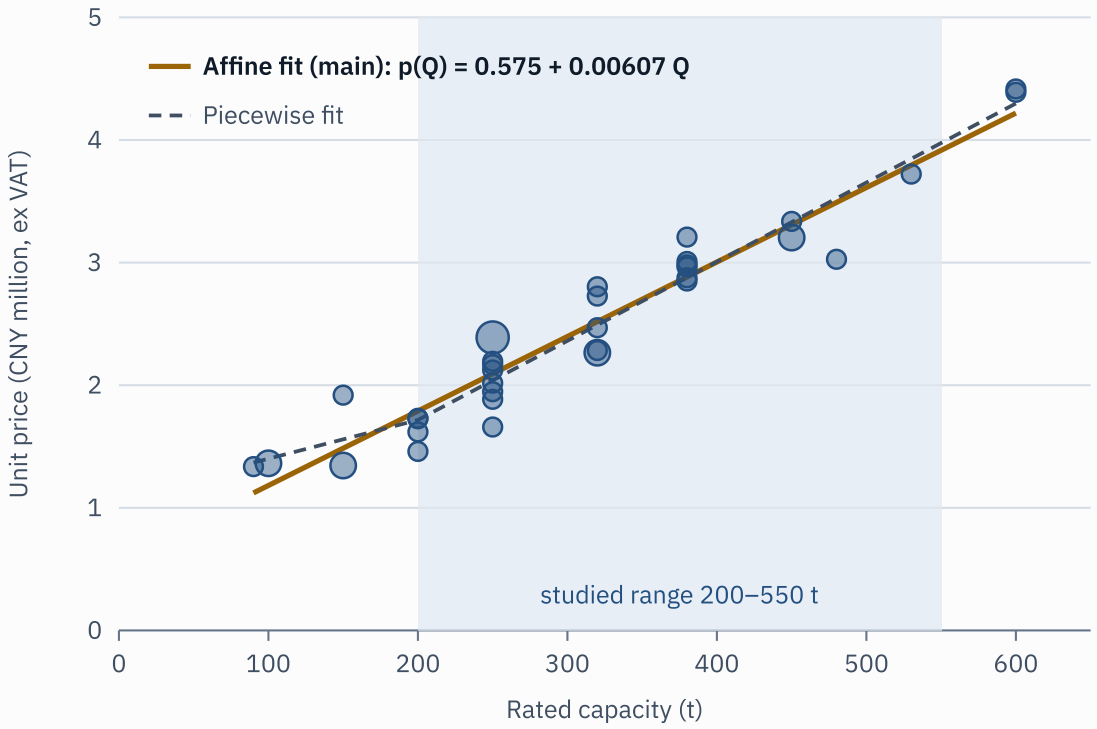
- In each scenario, the fleet with the lowest C_X is the **least-cost fleet**. The runner-up's excess over it is the **margin**.
- The last three measures re-price **the same schedules**. Schedules are not re-optimised for each.
- **Crews**: every transporter has its own crew of four, one driver and three guiders. Crews are not shared. Loading and unloading count as crew time. Shop and stockyard staff outside the crews are not counted.

The 6,480 decisions re-price the same 384 sized fleets. Only the cost parameters change, so they are **not independent experiments**. Shares of decisions are descriptive frequencies over the grid, not probabilities.

Sources: crew of four, the lower end of four or five per transporter in practice. See Park & Seo (2012), p. 392. The overtime factor 1.5 is our setting. Measures are defined in paper §3.2. Scenarios and decisions are in §3.4.

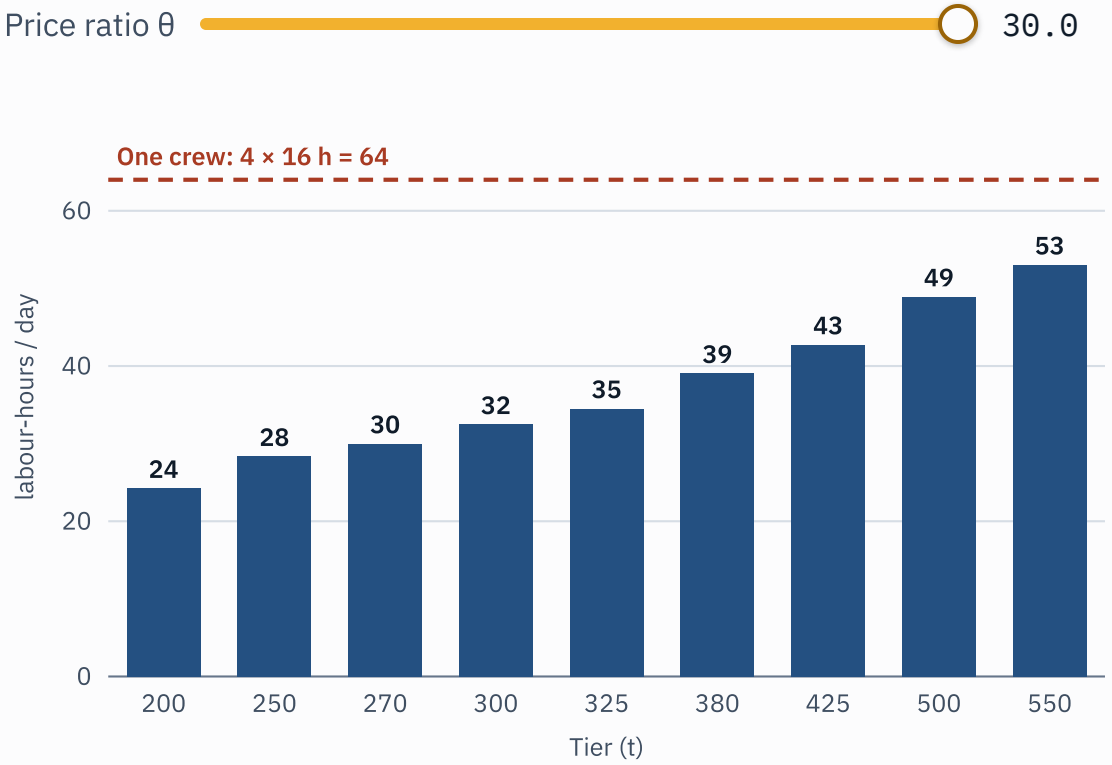
Prices: a vehicle-day of capital costs less than a crew-day

Unit prices of 39 transporters (33 award notices, 2022–2026, ex VAT). Shaded: studied range 200–550 t.



The 27 records within 200–550 t give α = 0.84 (95% CI 0.72–0.96). Over all 33 records, a lack-of-fit test gives p = 0.039. This argues against a single power law. So five price curves are used: affine (main), piecewise, and α = 0.72, 0.84, 0.96.

Per transporter-day: capital (bars) against one crew (dashed), in labour-hours

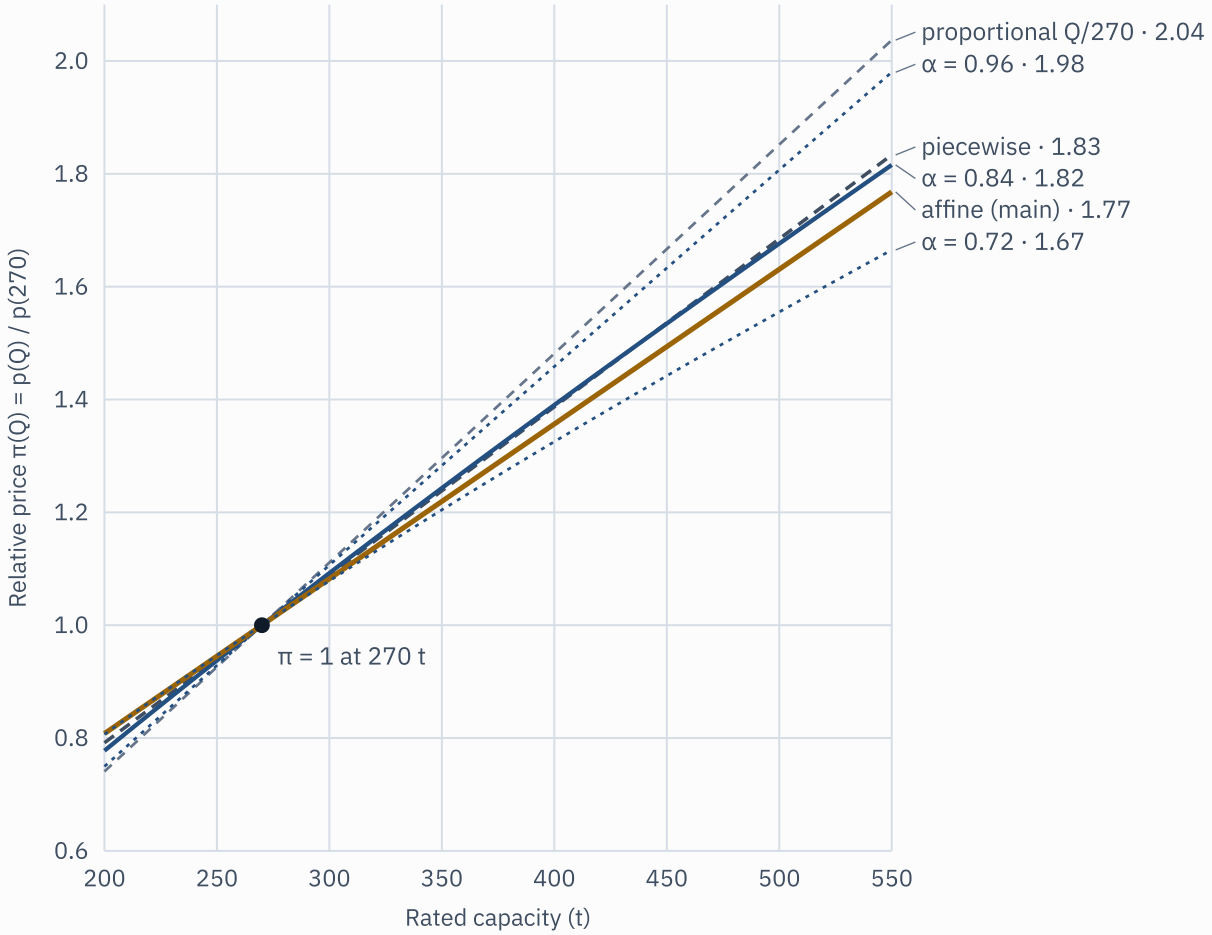


At θ = 30.0, the dearest tier, 550 t, costs 53 labour-hours of capital a day. That is still below one crew's 64. Saving a transporter beats moving up a tier.

Sources: 33 award notices, listed in Supplementary Table S4. They come from csscbidding.com, China Shipbuilding Electronic Bidding Platform (26), 365trade.com.cn (5) and e.sinochemitc.com (2). Fits and tests are our computation (paper §3.2, Table 3, Fig. 2a).

Above 270 t, all five price curves lie below proportional pricing

Price relative to a 270 t transporter: π(Q) = p(Q) / p(270). Range 200–550 t. Numbers on the right are the values at 550 t.



- **Affine** (main): p(Q) = CNY 0.575 million + 6,075 per t. **Piecewise**: CNY 1.085 million + 3,159 per t. Above 200 t, add 3,295 per t. Both are fitted to all 33 notices.
- **Power laws** $p \propto Q^{\alpha}$: the 27 notices within 200–550 t give α = 0.84. Its 95% CI is 0.72–0.96. A bootstrap over 20 buyers gives 0.73–0.95. The curves at 0.72 and 0.96 bound the interval.
- One power law fits all 33 notices poorly. Lack of fit: F(9, 22) = 2.49, p = 0.039. Within 200–550 t no lack of fit is detected: F(5, 20) = 0.86, p = 0.52.

> **One large unit costs no more than two smaller ones together. This is sub-additivity. Single-carrier dominance rests on it.**

Sources: curves fitted to the 33 award notices, ex VAT (Supplementary Table S4). Our computation (paper §3.2, Table 3, Fig. 2b).

Single-carrier dominance: one large-enough unit costs less than any coupled team

One 420 t block. Affine prices, θ = 12.4, shift labour, short handling.

	ONE 425 T ALONE	TWO 270 T COUPLED
Capital π	1.43	1 + 1 = 2.00
Crews	1	2
Occupancy	43 min	2 × (43 + 10) min
Cost per block	**3.7 labour-h**	**8.4 labour-h**

$$\text{cost per block} = (w_i + \delta \cdot 1[|S| > 1]) \div H \times \Sigma_{k \in S}\, \gamma(Q_k),$$
$$\gamma(Q) = \theta \cdot \pi(Q) + \ell$$

Fluid approximation: transporters are divisible and usable all day. Only the vehicle-time each block takes counts. Due dates, integer counts and schedule geometry are ignored.

Proposition 1: let prices be sub-additive, π(a + b) ≤ π(a) + π(b). All five price curves satisfy this over 200–1,650 t. Let the menu have a tier Q* with $m_i \le Q^*$ ≤ the team's capacity. Then one transporter of Q* carries block i at a lower fluid cost. The cost is strictly below that of any minimal team of g ≥ 2.

- **No more capital**: one larger unit costs no more than the members together
- **g – 1 fewer crews**: crews are paid per vehicle
- **No alignment**: every member is held δ longer

It holds with labour alone (θ = 0). It holds with capital alone (ℓ = 0, δ > 0). Even at δ = 0, a coupled lift adds (|S| – 1)·w + |S|·δ of transporter-time.

> **Corollary 1: suppose every block has a just-large-enough tier. Then the fluid-optimal fleet couples nothing and is mass-matched.**

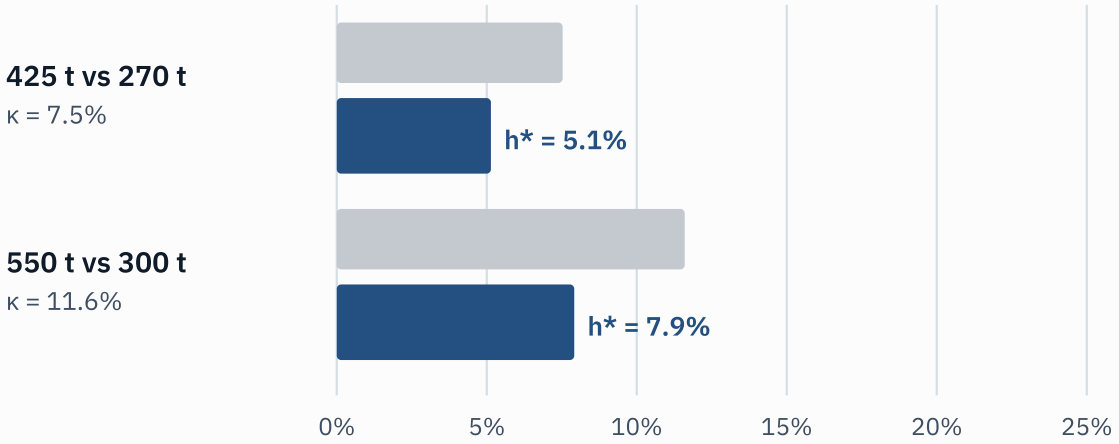
Sources: prices are five curves fitted to 33 tender awards. The 43 min is the covering fleets' mean occupancy per block with short handling (this study). The 10-min alignment is assumed. Paper §3.3, Assumption A, Proposition 1, Corollary 1.

Coupling budget: how much workload can a light tier couple and still be cheaper?

Price ratio θ 12.4

Alignment δ (min) 10.0

Occupancy w̄ (min) 43



κ: how much dearer the larger unit is per day

h*: the largest coupled share of workload the light tier can afford

h* = κ ÷ (1 + 2δ / w̄) = κ ÷ 1.47. δ = 10.0 min, w̄ = 43 min. w̄ is the covering fleet's mean occupancy per block. It is about 43 with short handling, 63–85 with mass-dependent handling and about 80 with long loading. Prices follow the α = 0.84 power law; labour is shift labour.

$$h \cdot (1 + 2\delta / \bar{w}) < \kappa \Leftrightarrow h < h^{*} = \kappa / (1 + 2\delta / \bar{w})$$

- **h**: the coupled share of single-carrier occupancy. It counts workload, not blocks. **κ** = $\gamma(Q_{hi}) / \gamma(Q_{lo}) - 1$: the covering tier's daily premium.
- Labour dilutes the capacity premium. For θ = 5.1–30, κ is only 3.4–14.8% for 425 vs 270 t. It is 5.3–22.5% for 550 vs 300 t.
- Covering fleets occupy a transporter about 43 min per block with short handling. It is 63–85 min with mass-dependent handling and about 80 min with long loading. With δ = 10 min the factor is 1.47 or 1.24–1.32.
- So **h* ≈ 2–18%** under shift labour and the fitted power law. Across the five price curves it is 1.9–21.8%. Yet in the 36 conditions the 200–380 t tiers couple **5–100%** of blocks.

The budget is small because labour dilutes the capacity premium.

Sources: κ uses the power law (α = 0.84) and shift labour. The power law is fitted to 200–550 t tender prices. θ = 5.1–30. Occupancy per block and 5–100%: this study (paper §5.1, §5.2). The 10-min alignment is assumed. Paper §3.3, Corollary 2.

What the model leaves out, and four implications to test

In real operations, not in the fluid model. Each can favour coupling.

Menu gap

If no tier can carry a block, coupling is necessary.

Integrality

Transporters are indivisible. A heavy tier kept for a thin heavy tail may stand idle. Light units with one or two heavy units are the integer counterpart of a mass-matched fleet.

Timing

Release peaks and due dates need slack beyond average workload. A coupled lift needs several vehicles free at once.

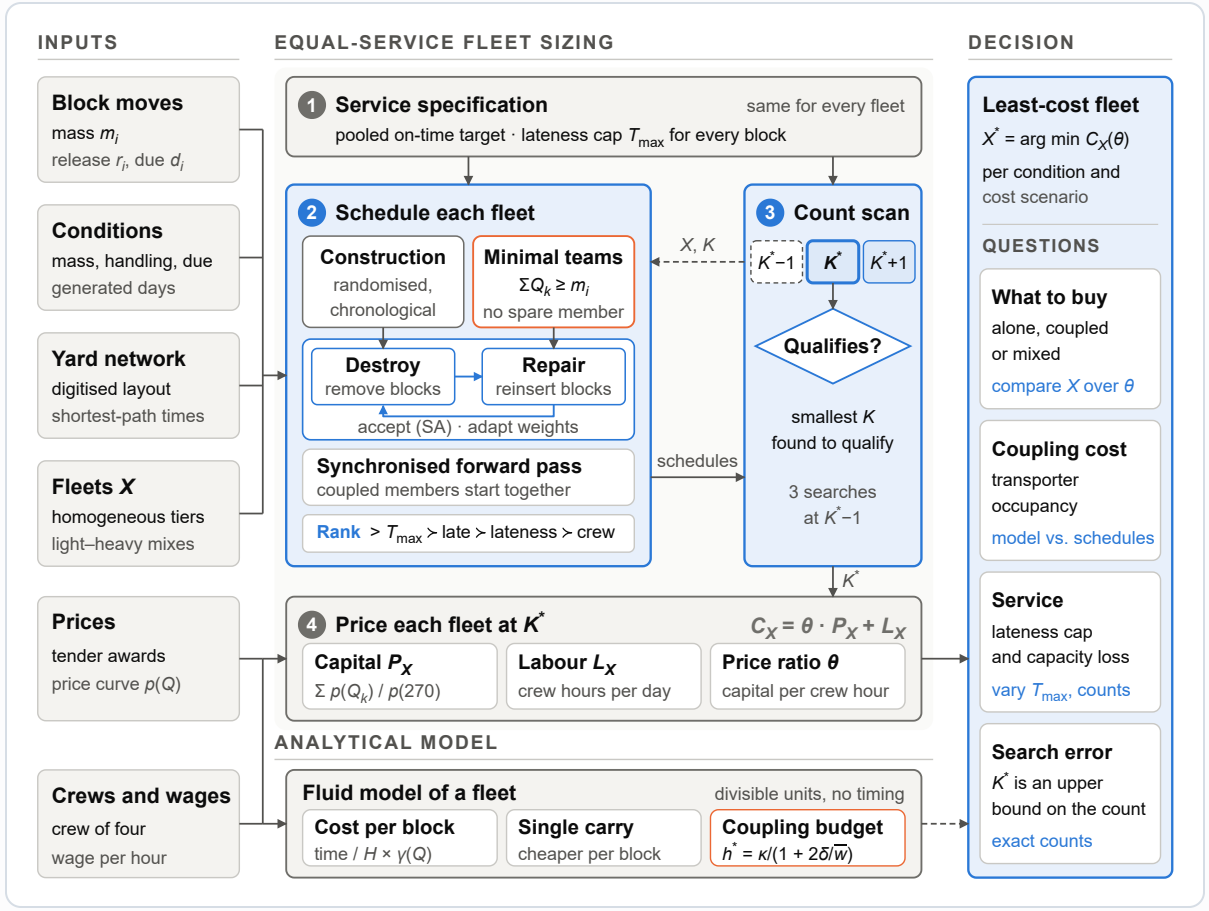
Four implications

1. Where some tier carries every block, least-cost fleets couple little
   → Results: what to buy
2. A light tier beats the covering tier only when its coupled occupancy is below h*
   → Why coupling costs: the calibrated screen
3. The heavy units of the cheapest light–heavy mix track the heavy blocks' share of occupancy
   → Composition: the heavy-share rule
4. Coupling-reliant fleets can win only where the budget does not bind or is large. Examples: a menu gap, near-zero alignment, or capital dear relative to labour.
   → Capital against labour; when coupling pays

Two kinds of heavy must be kept apart. Some blocks exceed a light tier but not the largest. Buying capacity serves them, so they govern the purchase. Other blocks exceed every tier. Every fleet must couple them, so they only set a floor on coupling.

Sources: paper §3.3.

Evaluation: specify service, schedule each fleet, scan the count, price at K*



- ① **Service specification**: the same for every fleet. At least 95% on time, pooled over 30 days. Every block within T_{max}.
- ② **Schedule**: ALNS handles single and coupled lifts. Rank by over $T_{max} \succ$ late $\succ$ lateness $\succ$ crew time.
- ③ **Count scan**: find the smallest qualifying K. Three searches a day at K* − 1. A schedule for K stays feasible for K + 1.
- ④ **Price**: $C_X = \theta \cdot P_X + L_X$. Each cost scenario picks the least-cost fleet.

Search error enters at step ③, where K* is an upper bound. It can change the decision at step ④. The checks of the findings test both. All fleets face the same blocks, locations and start points (common random numbers). A fleet of K uses the first K of 40 start points.

Sources: evaluation procedure of this study (paper §3.4, Fig. 4). Drawn in the colours of the paper's figures.

Case: 36 conditions on a digitised yard, plus a second yard from published data

Eight flows and their relative weights (assumed)

Assembly shop → Outfitting shop	0.5
Assembly shop → Paint shop	0.5
Outfitting shop → Paint shop	0.5
Paint shop → Block yard	0.7
Paint shop → Assembly yard	0.3
Block yard → Assembly yard	0.7
Assembly yard → Dock	0.2
Assembly shop → Assembly yard	0.05

Road network: the digitised layout of a large shipyard (not released). Travel times are static shortest paths, without traffic. Speeds: 12 km/h empty, 6 km/h loaded (270 t data sheet). Pickup and drop-off points are random within zones. Eight batches a day, one every 2 h. Each transporter starts at a random shop entrance. Process routes (shops, stockyard, dock) and road paths are inputs. They are not optimised, so every fleet serves the same demand.

36 × 30

6 mass profiles × 3 handling regimes × 2 due-date regimes. Each has 30 days: 1,080 days, 104,760 block moves.

11

reference fleet types: 9 tiers (200–550 t), and 270 t units with one or two heavy units. Heavy units are scheduled as 550 t. They are priced at the smallest tier that carries every block they carry alone.

384 · 6,480 sized fleets (200 and 250 t not run for the heavy tail) · decisions

214,210

scheduling runs (51,480 in the main experiment), about 2,570 core-hours. They ran on an Intel Core Ultra 9 275HX (24 cores) with 32 GB RAM. The system was Windows 11 with PyPy 3.9. Each run is seeded and budgeted by iterations, so it repeats exactly.

50

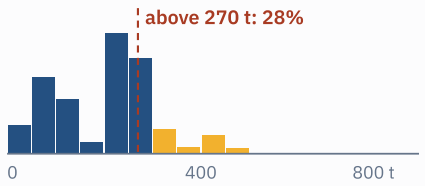
second yard: a Shanghai cruise-ship yard. Published data: a week of transfers, a distance table and five units (250–420 t).

Sources: 97 blocks a day Roh & Cha (2011). Speeds: Suzhou Dafang DCY270 data sheet. Flow categories and weights are assumed. Second-yard data: Liu et al. (2022), Tables 4–5 and supplementary data S1. Run counts are ours (paper §3, §4.1, §4.3).

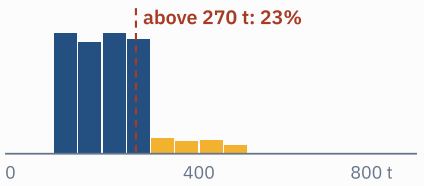
36 conditions = 6 mass profiles × 3 handling regimes × 2 due-date regimes

Six mass profiles: masses of the 2,910 blocks of 30 days. Orange: above 270 t.

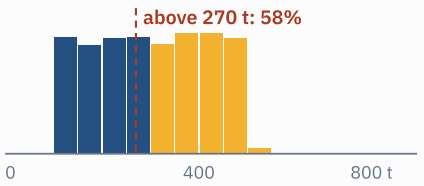
Jiang · 38–450 t
68 masses from the three instances of Jiang et al. (2021), resampled



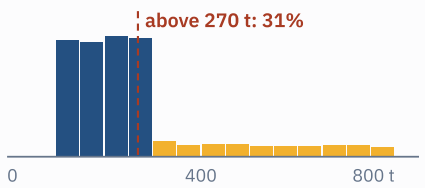
Light-skewed · 100–500 t
90% at 100–300 t (Park & Seo 2012), 10% at 300–500 t (assumed)



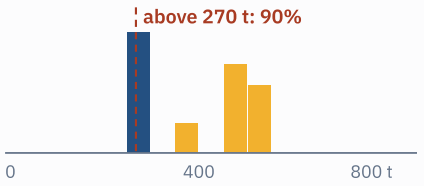
Uniform · 100–500 t
Uniform 100–500 t (the rule of the public instances of Kweon et al. 2026)



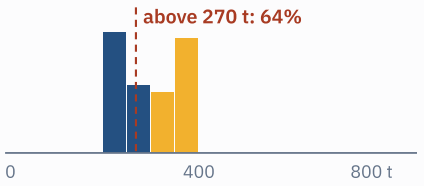
Heavy tail · 100–800 t
Light-skewed plus a 10% tail at 500–800 t: some blocks exceed every tier (assumed, a stress test)



Roh–Cha · 260–544 t
The ten block masses of Roh & Cha (2011) Table 2, resampled



Liu · 204–399 t
The 50 block masses of one week in Liu et al. (2022), resampled



Three handling regimes

- **Short**: 10 min each (Liu et al., 2022)
- **Mass-dependent**: U(10, 20) + 0.05 × mass − 5 min each (Kim & Joo, 2012)
- **Long loading**: 30–60 min loading, 10 min unloading (Liu et al., 2022)

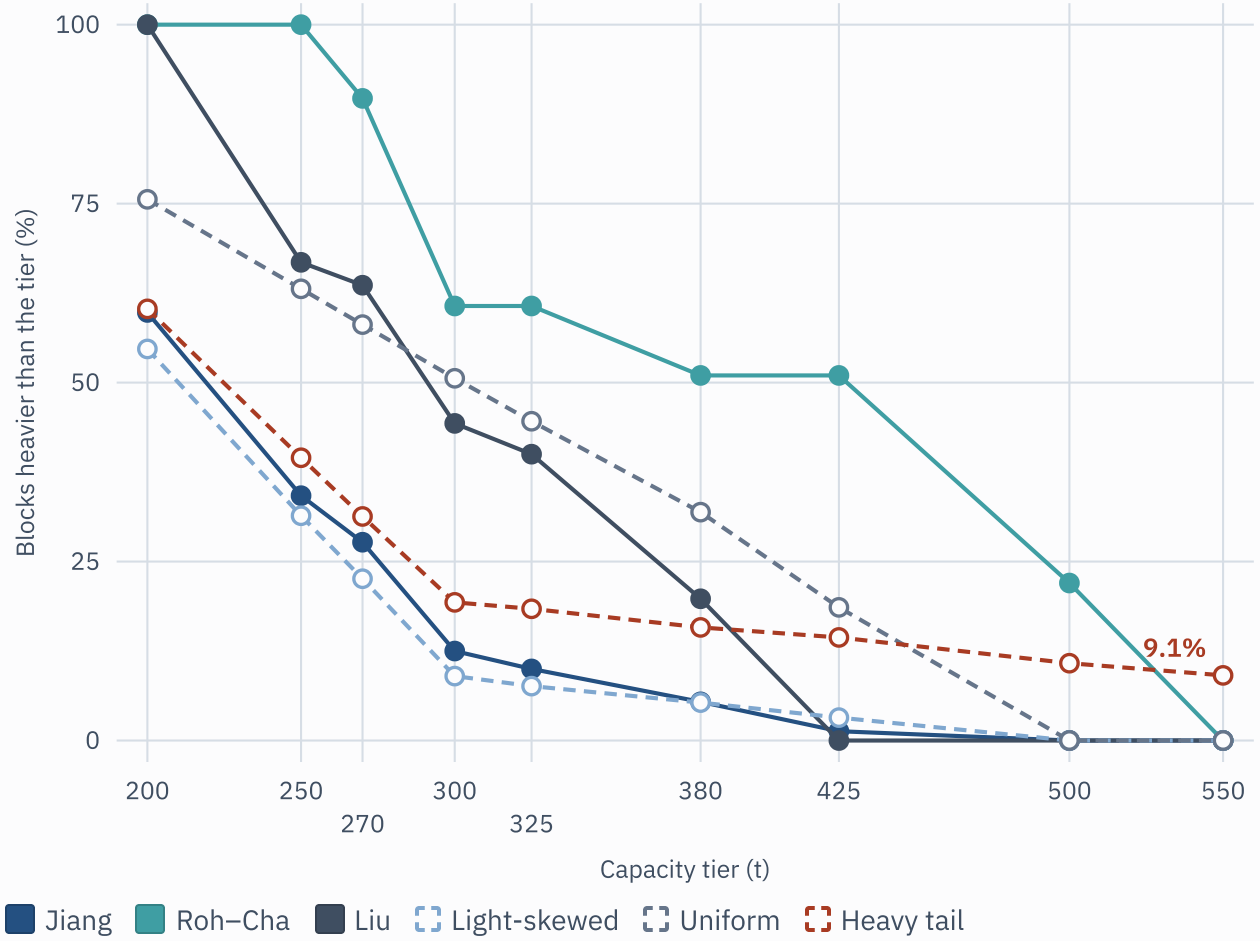
Two due-date regimes

- **Empirical**: 150 / 180 / 210 min. Probabilities 0.6 / 0.2 / 0.2 (Roh & Cha, 2011).
- **Tight**: 120–150 min. Assumed. It is tighter than the shortest published 150 min.

Sources: masses Jiang et al. (2021) Tables 5, A1, A2 (68 after removing repeats). Park & Seo (2012), p. 390 (100–300 t). Kweon et al. (2026) public instances (100–500 t). Roh & Cha (2011) Table 2. Liu et al. (2022) S1 data. The 10% at 300–500 t in the light-skewed profile is assumed. So is the 500–800 t tail. Handling: Liu et al. (2022) §4.1.1, §4.2.2. Mass-dependent formula: Kim & Joo (2012) §4. Due dates: Roh & Cha (2011) Table 2. Paper §4.1, Table 4, Fig. 5a.

Blocks heavier than a tier: the share its homogeneous fleet must couple

Share of blocks heavier than each capacity tier, 2,910 blocks per profile. Solid: published samples. Dashed: stylised profiles.



- The **covering tier** is the smallest tier that carries every block alone. It is 500 t for Jiang, light-skewed and uniform. It is 550 t for Roh–Cha and 425 t for Liu.
- The **heavy tail** has none. There, 9.1% of blocks exceed 550 t. Any fleet must couple them.
- Only 1.3% of the Jiang blocks exceed 425 t. Of the Liu blocks, 44% exceed 300 t.
- So across the 36 conditions the 200–380 t tiers couple 5–100% of blocks.

Below the covering tier the masses, not the schedule, fix how much a fleet couples.

Sources: the 2,910 block masses of each profile in the main experiment, released with the paper. This study (paper §4.1, Fig. 5b).

Parameter sources (1): values from the literature or public data

PARAMETER	VALUE	SOURCE (PAGE / TABLE)
Moves per day · shift	97 · 16 h	Roh & Cha (2011), p. 3246
Baseline due times	150 / 180 / 210 min (0.6 / 0.2 / 0.2)	Roh & Cha (2011) Table 2
Mass samples	38–450 t · 260–544 t · 204–399 t	Jiang et al. (2021) Tables 5/A1/A2. Roh & Cha (2011) Table 2. Liu et al. (2022) supplementary data.
Uniform masses · footprint	100–500 t · 5–15 m by 10–30 m	Kweon et al. (2026) public instances
Handling	10 min each · mass-dependent · 30–60 min loading	Liu et al. (2022) §4.1.1, §4.2.2. Kim & Joo (2012) §4.
Speed	12 empty · 6 loaded km/h. By model 10/5, 12/5.	Suzhou Dafang DCY270, DCY380, DCY500 data sheets
Tiers	200–550 t	Jiang et al. (2021) Table 5. Roh & Cha (2011) Table 1. Liu et al. (2022) Table 4.
Crew per vehicle	four: one driver and three guiders, not shared. The lower end of the published four or five.	Park & Seo (2012), p. 392
Prices	39 units, CNY 1.34–4.42 M, 90–600 t	33 award notices, 2022–2026. csscbidding.com: 26. 365trade.com.cn: 5. e.sinochemitc.com: 2. Supplementary Table S4.
Wages	National: CNY 79,182 a year (2025). Shanghai: CNY 184,833 (2024).	National Bureau of Statistics of China (2026), Table 7. China Statistical Yearbook 2025, table 4-12.
On-cost	30–40%	official contribution rates, ≤ 39.4% in total

Sources: each entry was checked against the original page and sentence. Full references are on the last slides.

Parameter sources (2): values we had to set, and how each was tested

PARAMETER	SETTING	TESTED AT
Release	eight batches, 2 h apart	identical for all fleets
Tight due dates	U(120, 150) min	a condition beside the published due dates
Heavy part of light-skewed · heavy tail	10% × 300–500 t · 10% × 500–800 t	Stress conditions. The heavy tail is analysed separately.
Alignment δ	10 min	0, 2.5, 5, 20, 30, 40 min. Low-coupling share 88.6% at δ = 0. It is 100% at 20–40 min.
Team size · turns	≤ 3 · 0 s	Teams of at most two. Turns of 60° or more: 30 s each. Coupled teams: 30 or 60 s per turn.
One speed for all tiers	12 / 6 km/h	Manufacturer speeds 10/5, 12/5 km/h. Speed × 0.75, × 0.5. Liu's 50/30 m/min.
Handling multiple	× 1	× 0.5, 0.67, 1.5, 2. Plus ±50% random noise.
Service	95% · 120 min	Caps of 30, 60, 240, 480 min and none. On-time targets of 90% and 98%.
Capital recovery · hours	8–15 y, 4–8%, 250–300 days. 2,000 h a year.	θ reported over all of 5.1–30. Extended to 0.5–5,000.
Sampled days	30 per condition	2,000 bootstrap resamples. 30 fresh days.
Search budget	786 rounds. 5,000 constructions.	Counts after constructions, 100 and 300 rounds. Six more searches a day at K* − 1. Exact counts on small instances.

Sources: results of these tests are in two later parts. They are “Composition (i) and service (iii)” and “Checks of the findings”. See paper §5.3–5.5 and Supplementary S2–S5. Manufacturer speeds: Suzhou Dafang DCY270, DCY380, DCY500 data sheets. Speeds of 50/30 m/min: Liu et al. (2022) §4.1.1.

Confirmation experiments: new instances test the model, exact solutions check the counts

Fresh days

30 days never used elsewhere. They cover the 42 fleets contending for least cost in seven conditions.

Extended mixes

13 further conditions. 300 t units with covering heavy units (500, 550 or 425 t). Heavy units: $K_H^* - 2$ to $K_H^* + 2$.

Slow speed and long handling

425 t, 500 t and mixes are added at slow speeds and handling ×1.5, ×2. Unresolved fleets are scanned up to 40.

Capacity loss

One fewer transporter in each capacity class, starts reassigned. Redundancy is estimated from qualifying counts.

On-time target

the 42 fleets re-sized at 90% and 98%

Exact benchmark

80 single-batch slices of 12–13 blocks, from ten days in eight conditions. CP-SAT finds the fewest late blocks at each count. 70 sized fleets.

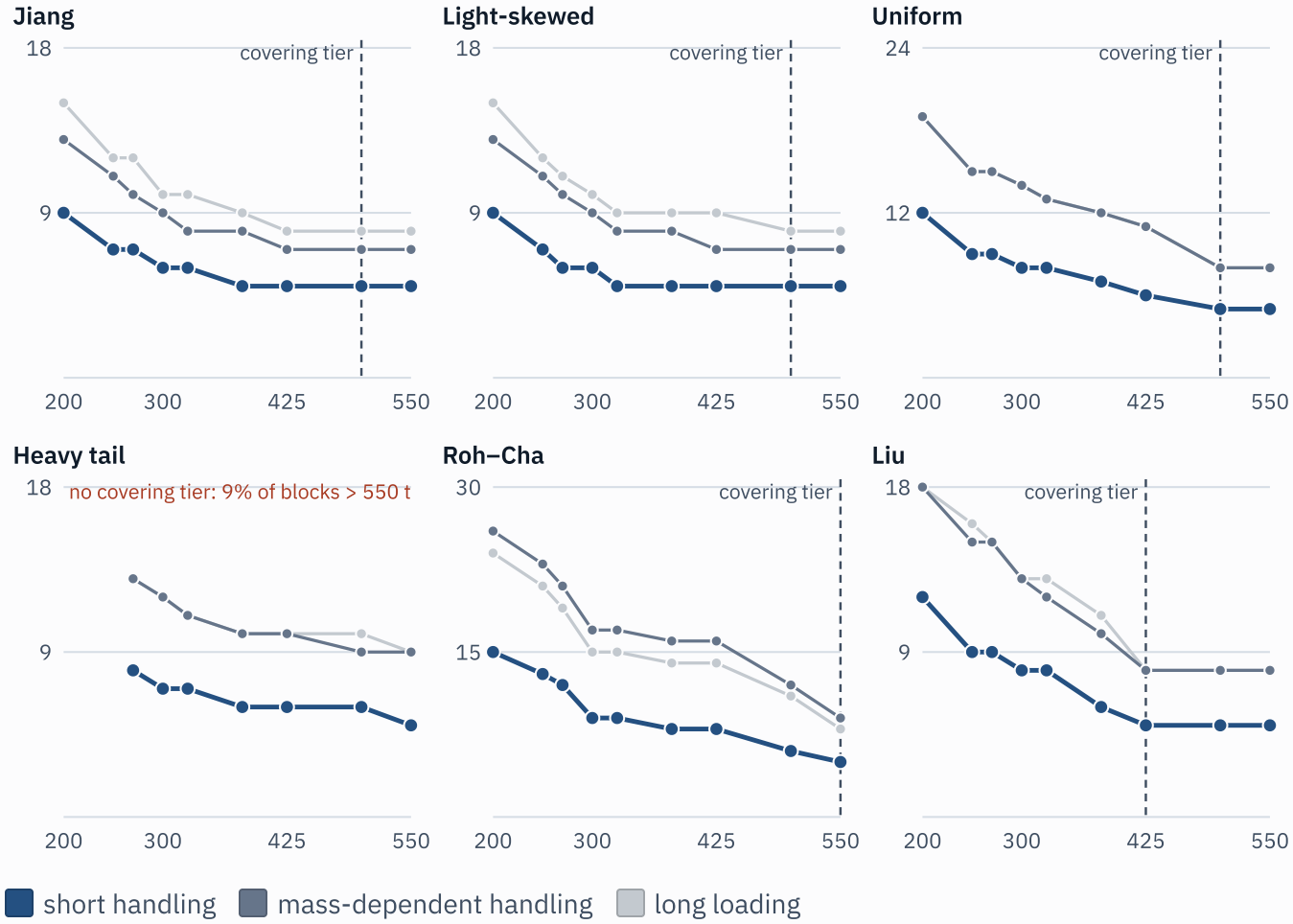
Heavy-share rule: $K_H^* = \lceil K_{cov} \cdot h_H \rceil$

K_{cov}: the count of the covering homogeneous tier. h_H: the share of handling time taken by blocks above 300 t. Three further analyses use the stored runs. They estimate redundancy from reduced compositions, re-select schedules per labour measure, and decompose transporter-time by block.

Sources: paper §4.2, Table 4. Supplementary S5, S7. Exact solves use CP-SAT of Google OR-Tools.

Question (i): larger tiers need fewer transporters, down to the covering tier

Smallest qualifying count K*, empirical due dates. Dashed: the covering tier, the smallest tier that carries every block alone.



- **Liu masses**, short handling: 12 units of 200 t and 9 of 270 t. Then 6 of 380 t and 5 of 425 t or more.
- **Roh–Cha masses**: 15, 12, 8 and 5 units of 200, 270, 425 and 550 t.
- Slower handling raises every line, but the shape stays. Under Roh–Cha masses, 200 t needs up to 26 units, 30 with tight due dates.
- Beyond the covering tier, extra capacity buys nothing. Under Liu masses, 425, 500 and 550 t need the same count in all six conditions.
- Below it, the masses fix the coupled share. The 200–380 t tiers couple 5–100% of blocks.

A light fleet's extra units are the price of coupling.

Sources: this study's count scans, 11 fleet types in 36 conditions. Paper §5.1, Fig. 6.

Question (i): the least-cost fleet carries almost every block alone

91.0%

of 1,620 shift-labour decisions: the least-cost fleet couples ≤ 1/10. All 1,350 with a covering tier do.

57.5%

of all 6,480 decisions: the winner couples no block

24.1%

won by light units with one or two heavy units, coupling ≤ 1/10. Each unit carries its own blocks.

1.2%

won by a light tier of 325 t or less. Only in the heavy tail or the hardest Jiang condition.

By labour measure (share coupling ≤ 1/10, 1,620 decisions each)



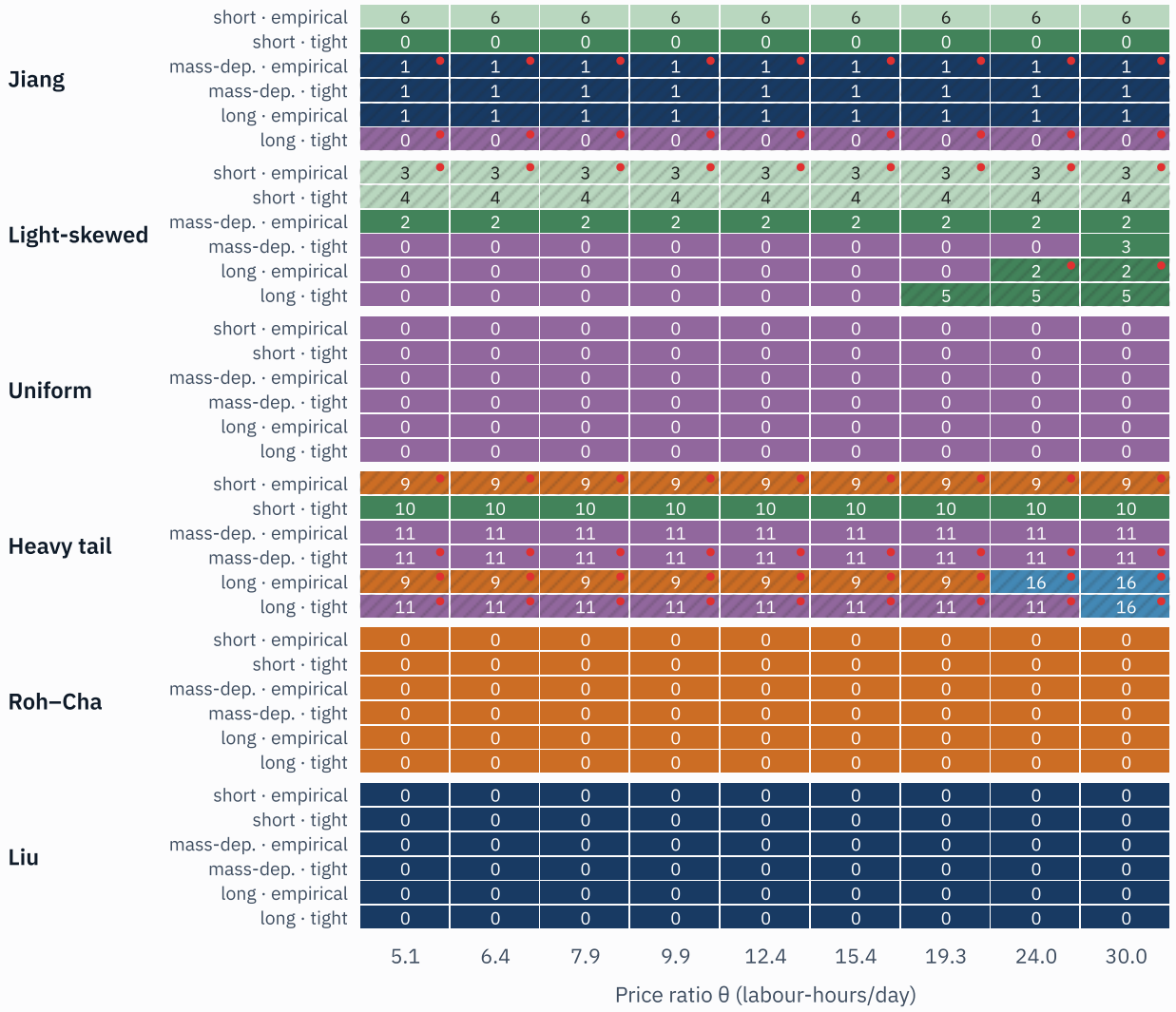
* One crew drives a coupled team: the measure most favourable to coupling. Outside the heavy tail, all 32 wins of coupling-reliant fleets use it.

- Over all four measures: 90.8% (5,882 of 6,480). Homogeneous fleets carrying alone win 66.7%.
- **The threshold is not critical**: ≤ 2.5% in 74.5%, ≤ 15% in 97.3%. Price curves: 90.0–91.6%. θ 5.1 → 30: 93.1% → 87.6%.

> **Every minimal team is assumed physically feasible. That favours coupling, so it cannot explain these shares.**

Sources: this study, 6,480 decisions (paper §5.1, Fig. 7). A decision = condition × price curve × θ × labour measure (36 × 5 × 9 × 4).

All 36 conditions × 9 price ratios: which fleet is cheapest



Shift labour | Crew-team hours

- **Colour**: the winning fleet. **Number**: the share of blocks it couples (%).
- **Hatched**: winner kept in < 80% of day resamples, a near tie.
- **Red dot**: a decision that more search might flip.

380 t · 425 t · 500 t · 550 t · 270 t + 1 heavy · 270 t + 2 heavy

support < 80% · weak search dependence

Example: the bottom six rows are the Liu masses. They are all dark blue: 425 t is cheapest at every price. The 0 means it never couples. Hover a cell for details.

Affine price curve. The row "heavy-tail · short handling · empirical due dates" has one known search miss. There, 270 t with two heavy units qualifies with one unit fewer. It is then cheaper than 550 t in 89 shift-labour decisions.

Sources: this study, 6,480 decisions (paper §5.1, §5.5).

By mass profile: the smallest near-covering tier, or light plus a few heavy units

MASS PROFILE	COVERING TIER	WINNERS (SHARE OF 1,080 DECISIONS)	COUPLES ≤ 1/10	COUPLES NONE	MEDIAN MARGIN
Jiang	500 t	425 t 42.1% · 270 t + 2 heavy 38.1% · 500 t 11.8% · 270 t + 1 heavy 5.6% · 300 t 2.2% · 380 t 0.2%	97.8%	11.8%	1.7%
Light-skewed	500 t	270 t + 2 heavy 59.2% · 500 t 24.2% · 270 t + 1 heavy 16.7%	100%	34.0%	1.6%
Uniform	500 t	500 t 99.3% · 270 t + 2 heavy 0.7%	99.3%	99.3%	2.0%
Heavy tail	–	270 t + 2 heavy 36.6% · 500 t 28.5% · 550 t 22.5% · 380 t 6.4% · 300 t 3.4% · 325 t 1.8% · 425 t 0.8%	47.6%	0%	1.2%
Roh–Cha	550 t	550 t 100%	100%	100%	30.1%
Liu	425 t	425 t 100%	100%	100%	3.3%

- In the five profiles with a covering tier, 99.4% of winners couple ≤ 1/10. Only 1.3% of Jiang blocks exceed 425 t, so 425 t wins.
- **The heavy tail is the expected exception** (a menu gap): 9.1% of blocks exceed every tier. Even 550 and 500 t couple 9.1% and 10.8%. Of 145 shift-labour decisions whose winner couples > 1/10, 126 go to 500 t. It couples only blocks above 500 t.
- **Margins are often small**: the runner-up costs a median 2.4% more, under 1% in 18.3% of decisions. Outside the heavy tail, fleets within 1% all couple ≤ 1/10 in 98.8% of decisions. Within 2%: 96.8%. The conclusions concern the kind of fleet, not one tonnage.
- Quotations: with other costs fixed, overturning the winner takes a median 11.1% capital-price rise. It takes under 5% in 1,123 decisions (17.3%). So purchases need comparable quotations.

Sources: this study (paper §5.1, Table 5, Supplementary S9). Mass-profile sources: “Parameter sources (1)”.

Try it: pick a condition and a price ratio, see the cheapest fleet

Masses: Uniform | Handling: short handling | Due dates: empirical due dates | Shift labour | Crew-vehicle hours | Crew-team hours

θ 12.4

Fleet	Composition	K*	Daily cost (labour-h)		vs cheapest	Coupled
500 t	5 × 500 t	**5**		**421**	★	0%
550 t	5 × 550 t	**5**		430	+2.0%	0%
270 t + 2 heavy	2 × 500 t + 4 × 270 t	**6**		474	+12.6%	11%
425 t	6 × 425 t	**6**		490	+16.4%	19%
270 t + 1 heavy	1 × 500 t + 6 × 270 t	**7**		543	+28.9%	33%
380 t	7 × 380 t	**7**		561	+33.2%	32%
300 t	8 × 300 t	**8**		619	+47.1%	51%
325 t	8 × 325 t	**8**		626	+48.7%	45%
250 t	9 × 250 t	**9**		681	+61.8%	63%
270 t	9 × 270 t	**9**		688	+63.3%	58%
200 t	12 × 200 t	**12**		888	+110.9%	76%

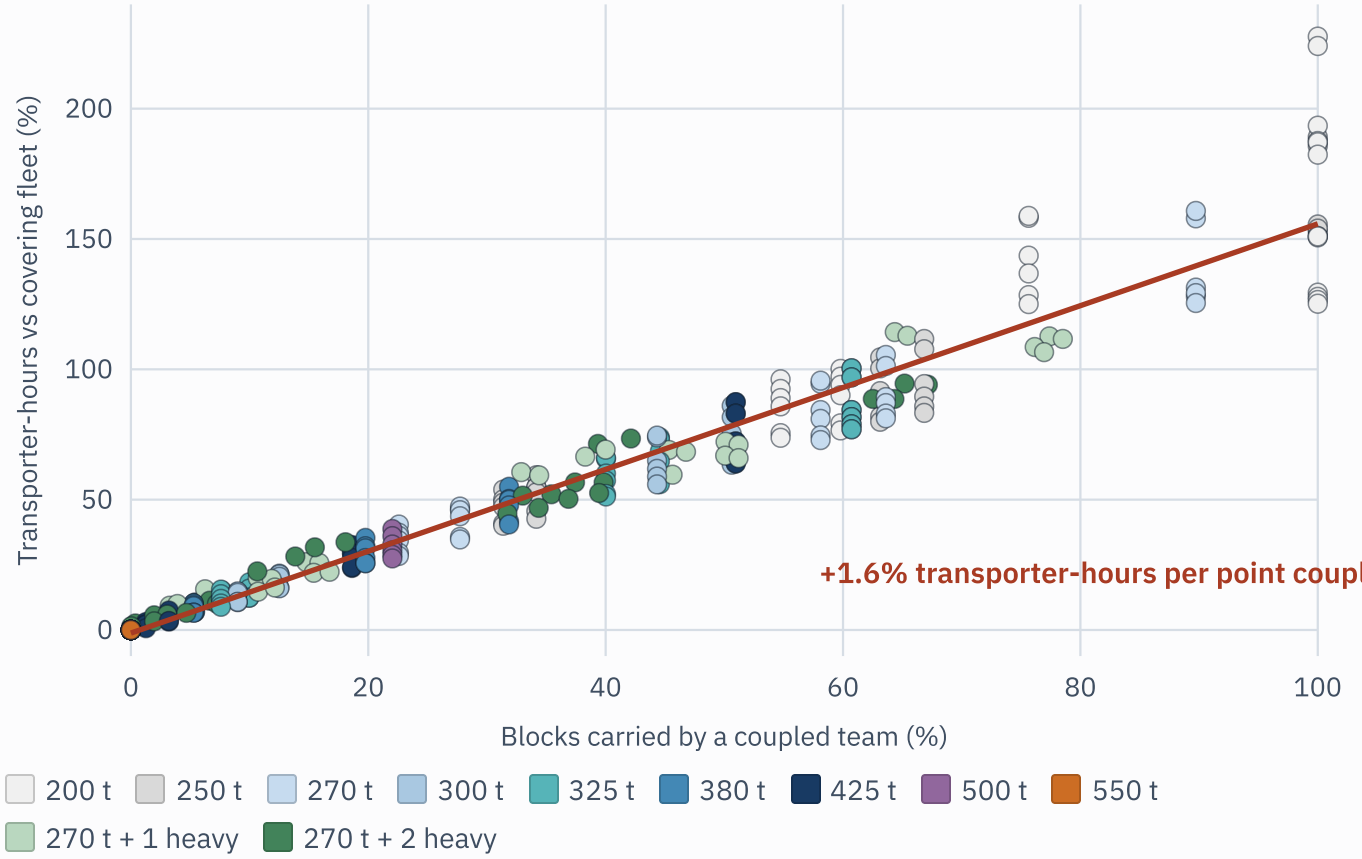
Cheapest: **500 t** (5 vehicles), 421 labour-hours a day. The runner-up, 550 t, costs 2.0% more. Labour is 76% of the cheapest fleet's cost. Capital θ × P Labour L

Daily cost C = θ × P + L, in labour-hours a day. P is the fleet price in 270 t equivalents (affine curve). Heavy units in mixes are priced at the smallest tier they need. Shift labour is 64 labour-hours per unit. Crew measures count only hours actually worked.

Sources: counts and hours, this study. Price curve fitted to 33 tender awards. 4 crew × 16 h: Park & Seo (2012), Roh & Cha (2011). Paper §3.2.

Question (ii): extra transporter-time comes from repeated service and alignment, not partner waiting

330 sized fleets outside the heavy-tail profile: transporter-hours a day above the covering fleet of the same condition



+1.6%

transporter-hours a day for each percentage point of blocks coupled (r = 0.97)

2.2×

the covering fleet's transporter-hours, for fleets that couple 60–100% of blocks

0.55%

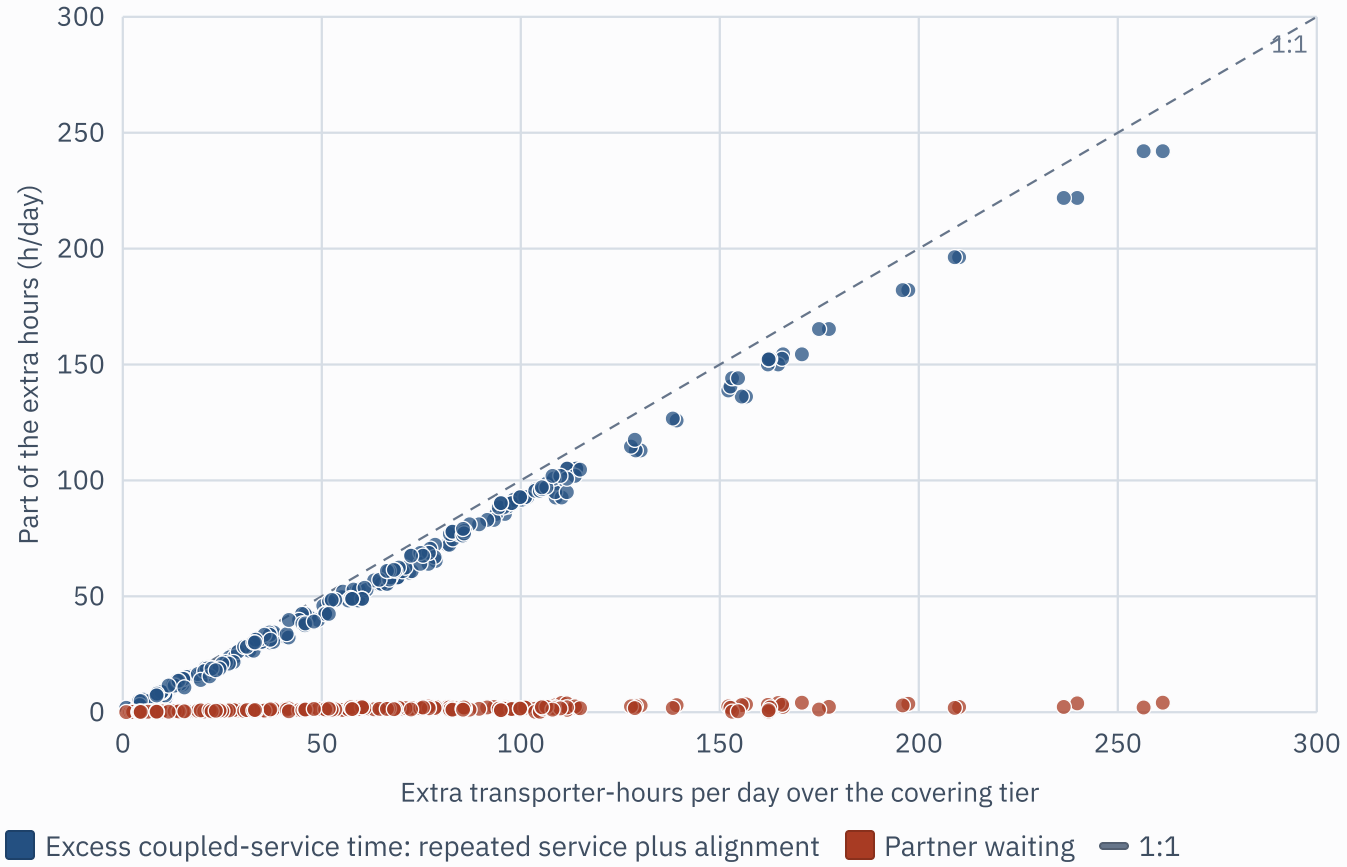
waiting for the last team member: a median 0.55% of working time. No fleet exceeds 2.2%.

Partner waiting is small. The cost: two or three vehicles and crews are held for the whole loaded trip, alignment included. They leave parallel service.

Sources: this study, working and waiting hours of each sized fleet (paper §5.2, Fig. 8a).

Block by block: repeated service and alignment are a median 90% of extra time

266 lighter fleets, each against the covering tier of its condition. Only fleets outside the heavy tail that couple > 1% of blocks. Shown: extra transporter-hours and their two parts.



89.8%

median share of the extra transporter-time that is excess coupled-service time. Per block: (|S| − 1)·D + |S|·δ, repeated service plus alignment. 10th–90th percentile: 77.5–93.8%.

2.1%

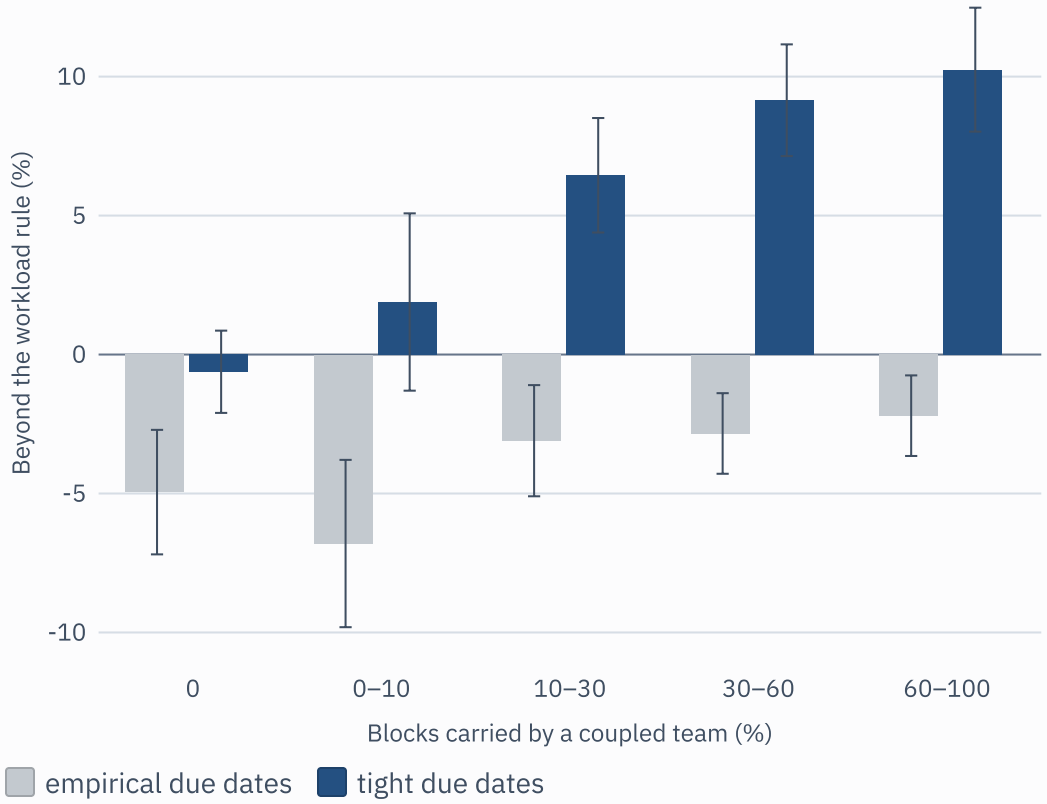
median share spent waiting for partners (1.3–3.2%). Early members could leave later, so this is an upper bound.

- The rest is the difference in empty travel, which can be negative. Points above the 1:1 line have less empty travel than the covering tier.
- The same decomposition gives the covering fleet's occupancy per block for the budget. It is about 43 min short, 63–85 mass-dependent, about 80 long loading.

Sources: this study, block-by-block decomposition of the priced schedules (paper §5.2, Fig. 8b). It covers lighter fleets among the 384 outside the heavy tail. They couple more than 1% of blocks.

Fluid screen: the lighter tier is ruled out in 77% of comparisons, once wrongly

Transporters needed beyond the workload rule K_p (share of K*, mean with 95% interval)



r = 0.97

fluid occupancy factor against the simulated count ratio K^*_{lo} / K^*_{hi} (210 fleet pairs)

77% · 1

with a 10% margin (fluid cost ratio > 1.10). The screen rules out the lighter tier in 7,311 of 9,450 comparisons. The simulation finds it cheaper only once, by 0.1%.

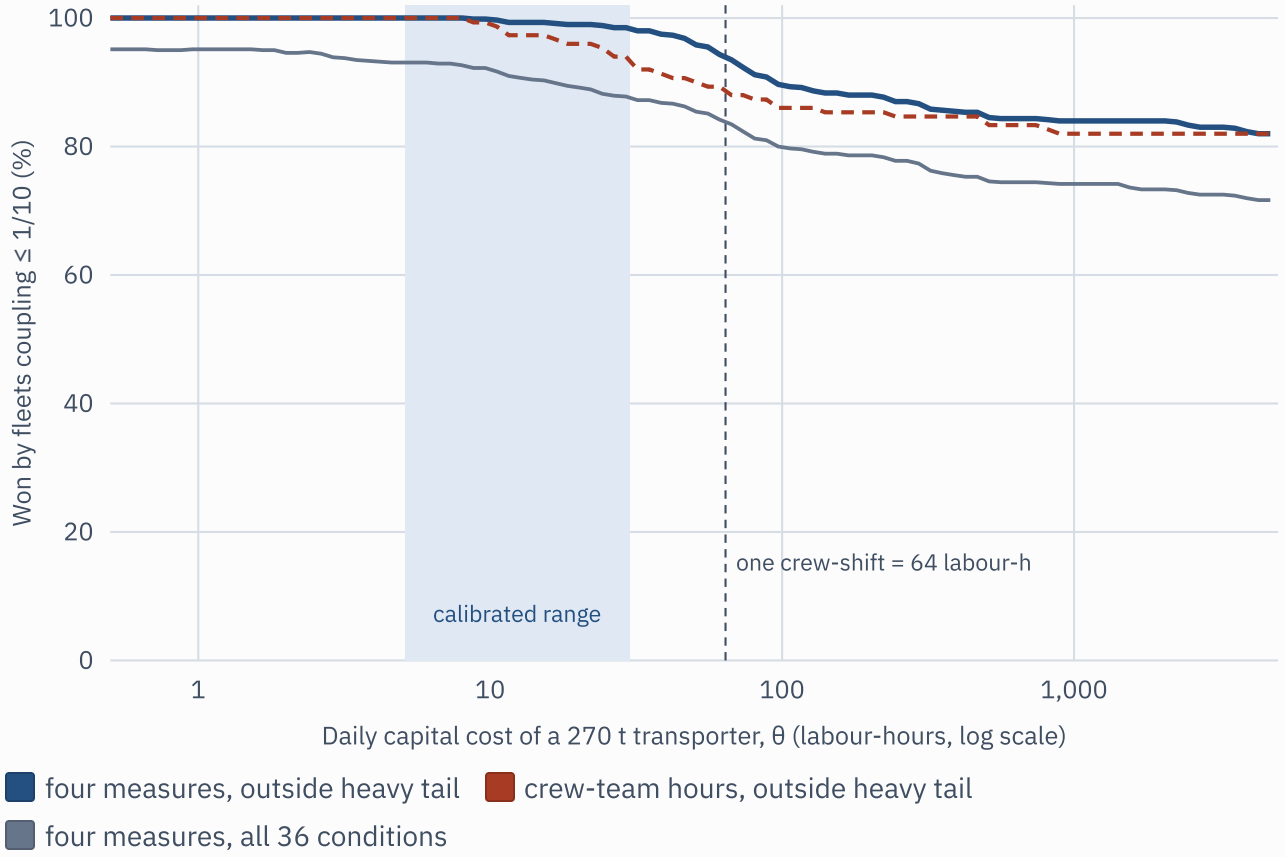
- The lighter tier wins 571 comparisons. 570 lie among the 2,139 left to scheduling. A 5% margin rules out 82.0% with 35 errors; no margin, 92.3% with 274.
- K_p is within one of K* for 319 of 384 fleets, and all 192 with empirical due dates. Under tight due dates, fleets coupling > 60% need 10.2% of K* more. Those coupling none need 0.6% fewer. A coupled lift needs several vehicles free at once.
- Priced on K_p: low-coupling share 92.0% (against 91.0%). But the same winner in only 57.7%. 17.7% of the selections lack a qualifying schedule.

The 77% is a within-sample check. Occupancy comes from covering-fleet schedules. A screen from inputs alone is untested.

Sources: this study (shift labour; paper §5.2, Fig. 8c; Supplementary S9). Each lighter homogeneous tier against the covering tier, in every condition with one. 5 price curves × 9 values of θ.

Coupling-reliant fleets could win only if capital became far dearer than crews

Decisions won by fleets coupling ≤ 1/10, for θ from 0.5 to 5,000. Lines pool conditions, 5 price curves and the measures named.



With counts and schedules fixed, every fleet's cost is a straight line in θ. So the lower envelope gives the winner at any θ exactly.

100% · 98.3%

outside the heavy tail: every decision up to θ ≈ 8.7. At θ = 30: 98.3%.

82%

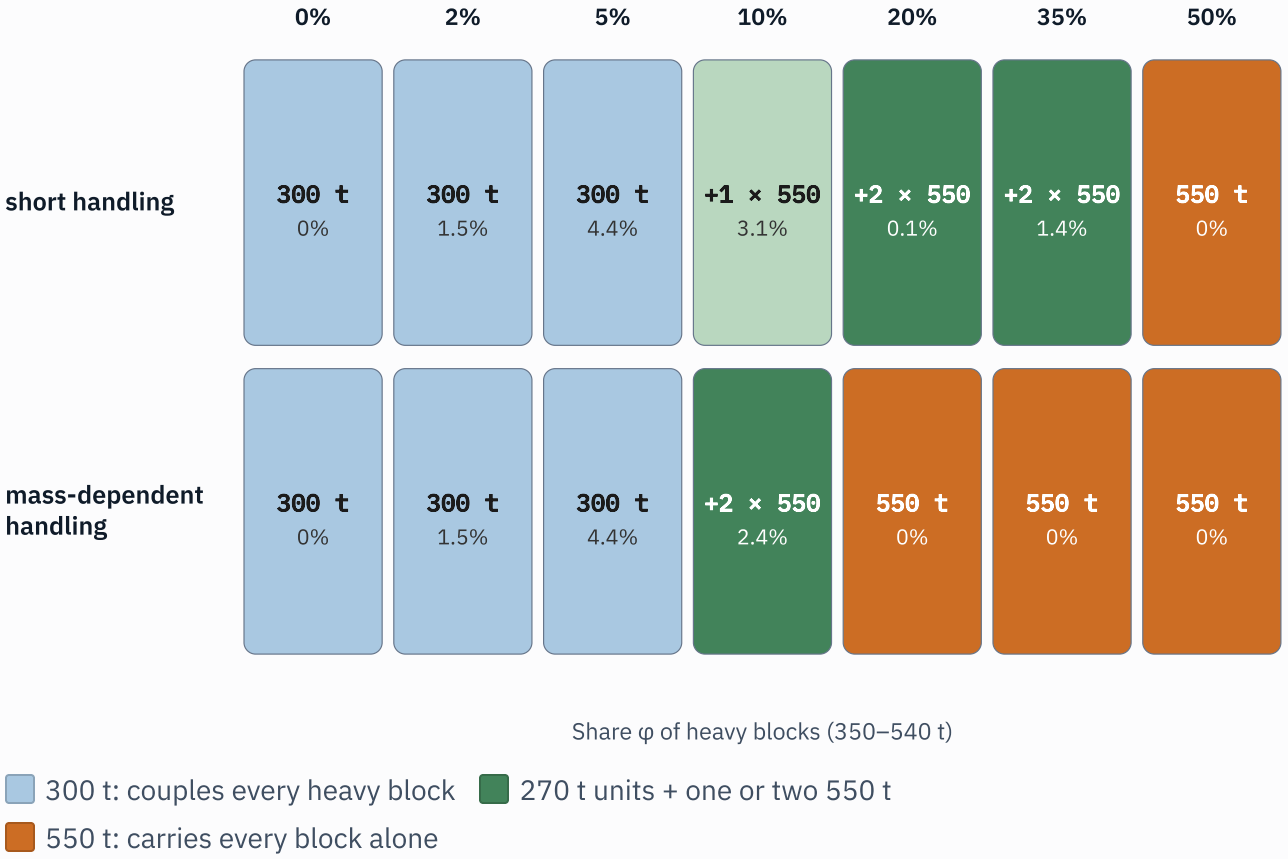
even with capital alone (θ → ∞). Sub-additive prices keep a single carrier cheaper per block (Proposition 1).

- In 492 of 600 combinations (82%), no θ up to 5,000 favours coupling-reliant fleets. Of the other 108, 98 switch above θ = 30 (median 87 labour-h, ~1.4 crew-days).
- Switches inside the calibrated range come only under crew-team hours (dashed). That measure assumes one crew drives a coupled team.
- A crew-day costs 2.1–12.5 times a 270 t unit's daily capital. Driverless transporters could approach the switch.

Sources: this study (paper §5.2, Fig. 8d). The calibrated range 5.1–30 comes from tender prices, official wages and assumed capital-recovery settings. See the two parameter slides.

Question (i): as heavy blocks grow, the cheapest fleet gets heavier but rarely couples

Cheapest fleet by the share φ of heavy blocks. Heavy blocks weigh 350–540 t, the rest 100–300 t. Affine price, shift labour. All nine price ratios give the same result.



- At φ = 0, 300 t wins every decision. At φ = 2%, it wins 92–97%. From φ = 5–10%, light units with one or two heavy units take over.
- 550 t first wins at φ = 50% (short handling) and φ = 20% (mass-dependent). There it wins all 45 shift-labour decisions. The true switch lies between neighbouring grid points.
- The coupling budget explains both switches. Up to φ ≈ 5%, 300 t stays within its budget. Once the heavy workload exceeds two heavy units, the mixes lose to 550 t.
- The simple formula ranks 300 t against 550 t correctly in 94.4% of 630 comparisons.

2,502 / 2,520

decisions where the cheapest fleet couples at most one block in ten. The exceptions are the two-heavy mix (19.6% coupled). They occur at φ = 35% with mass-dependent handling under crew-team hours.

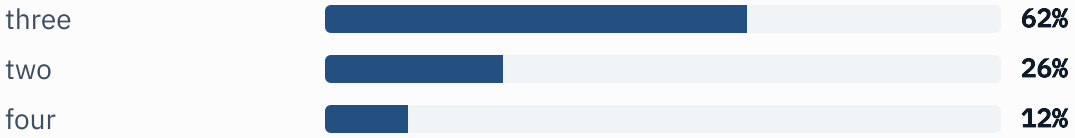
The winners' mean coupled share is at most 3.3% at every φ.

Sources: this study, 30 further days with φ from 0 to 50%. See paper §5.3 and Fig. 9. The ranges 350–540 t and 100–300 t are assumed.

Richer mixes change the cheapest fleet, not the principle

Liu sample, short handling, 120-min cap. Candidates are the 11 reference fleets plus extended mixes. 5 price curves × 400 values of θ × 4 labour measures = 8,000 cost scenarios. A mix of 300 t and 425 t units is cheapest in all of them.

Heavy 425 t units in the cheapest mix (share of scenarios)



44% of the Liu blocks exceed 300 t. A mass-matched five-vehicle fleet would hold about 2.2 heavy units. The cheapest mixes couple at most 2.1% of blocks.

0.3–7.7% is how much more the best other candidate costs

1.3–7.7%

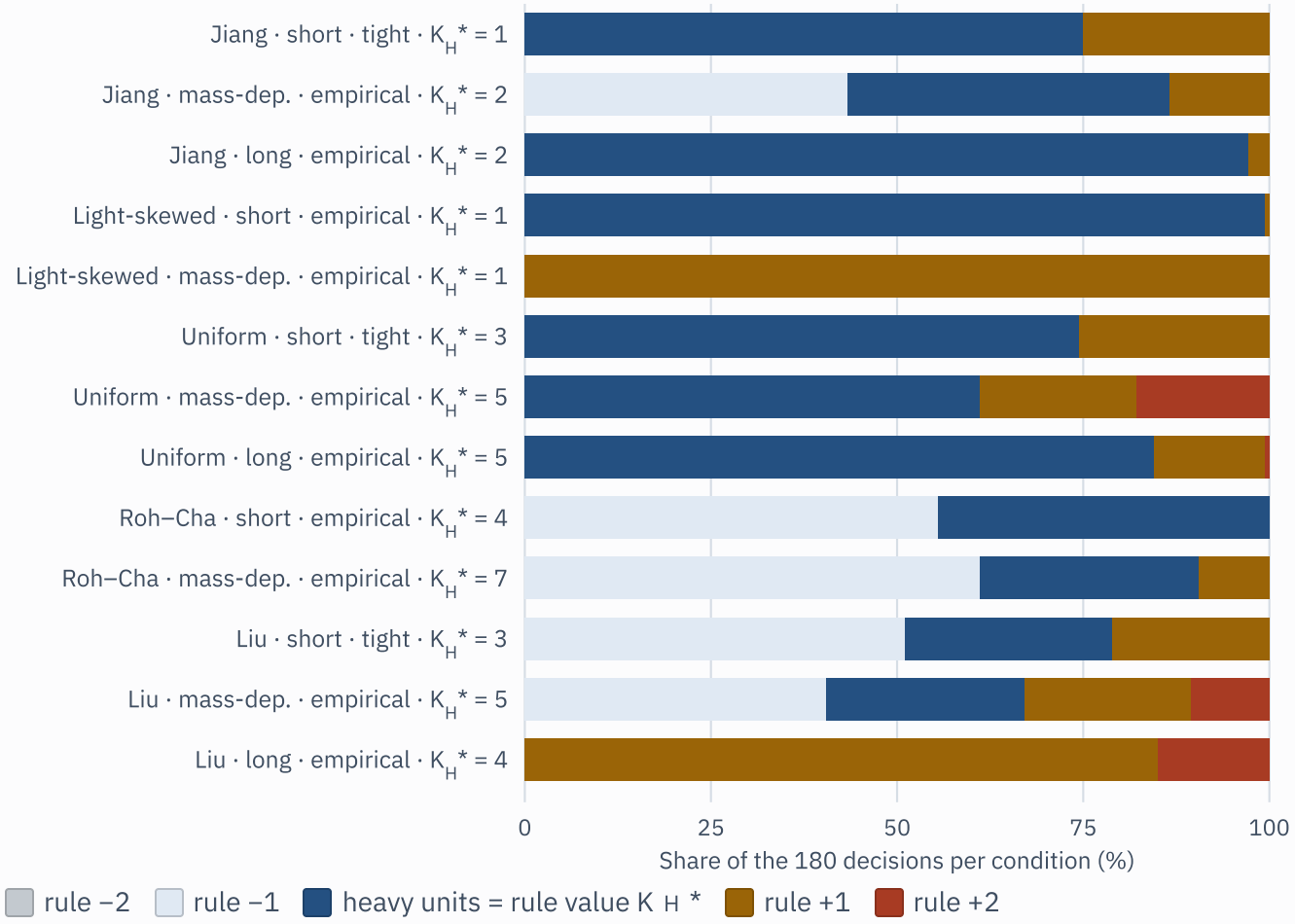
is how much more the homogeneous 425 t fleet costs under shift labour. It won among the reference types.

> **The number of heavy units follows the heavy workload. Richer mixes change which fleet is cheapest, not the single-carry principle.**

Sources: this study, the extended candidates of the lateness-cap study (paper §5.3). Liu masses come from Liu et al. (2022), S1 data.

In 13 further conditions, light units with matched heavy units win 95.3%

Heavy units of the cheapest 300 t mix, against the rule value $K_H^* = \lceil K_{cov} \cdot h_H \rceil$. 180 decisions per condition.



95.3%

of 2,340 decisions go to 300 t units with covering heavy units. These winners always couple ≤ 1/10. The best reference fleet costs a median 1.9% more, at most 10.2%. The exceptions are light-skewed. There 270 t units with one or two heavy units win, also light–heavy mixes.

96.6%

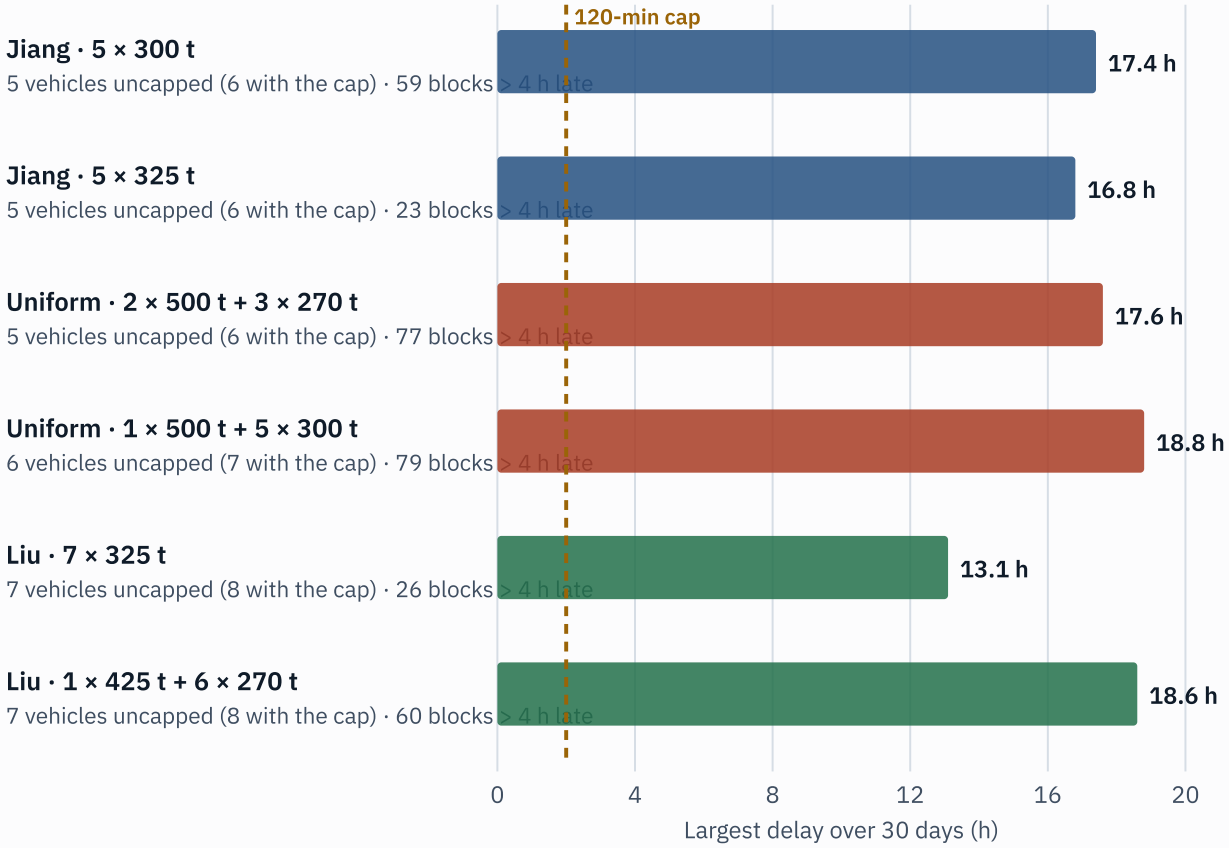
of decisions have a heavy count within one unit of K_H^*. The most frequent count equals K_H^* in 6 of the 13 conditions. In the other seven it is one off.

- Too few heavy units show the budget at work. Take the uniform profile with tight due dates. One 500 t unit leaves 27% of blocks coupled and needs seven transporters. Three need five and couple 0.3%.
- In four conditions, 79 decisions (3.4%) pick $K_H^* + 2$, the largest count tested. In three, one more unit gives the covering fleet, already a candidate. Only the Liu sample with long loading (1.2%) could hide a cheaper count.

Sources: this study, 13 further conditions (paper §5.3; Supplementary S5). 300 t units have $K_H^* - 2$ to $K_H^* + 2$ covering heavy units. These weigh 500, 550 or 425 t.

Question (iii): with an on-time share alone, fleets buy less by deferring blocks

These fleets need one unit fewer without a cap. Shown: the latest block of each over 30 days.



6

sized fleets qualify with one unit fewer without a cap. They are five fleet types in three conditions. Their latest block is 13–19 h late.

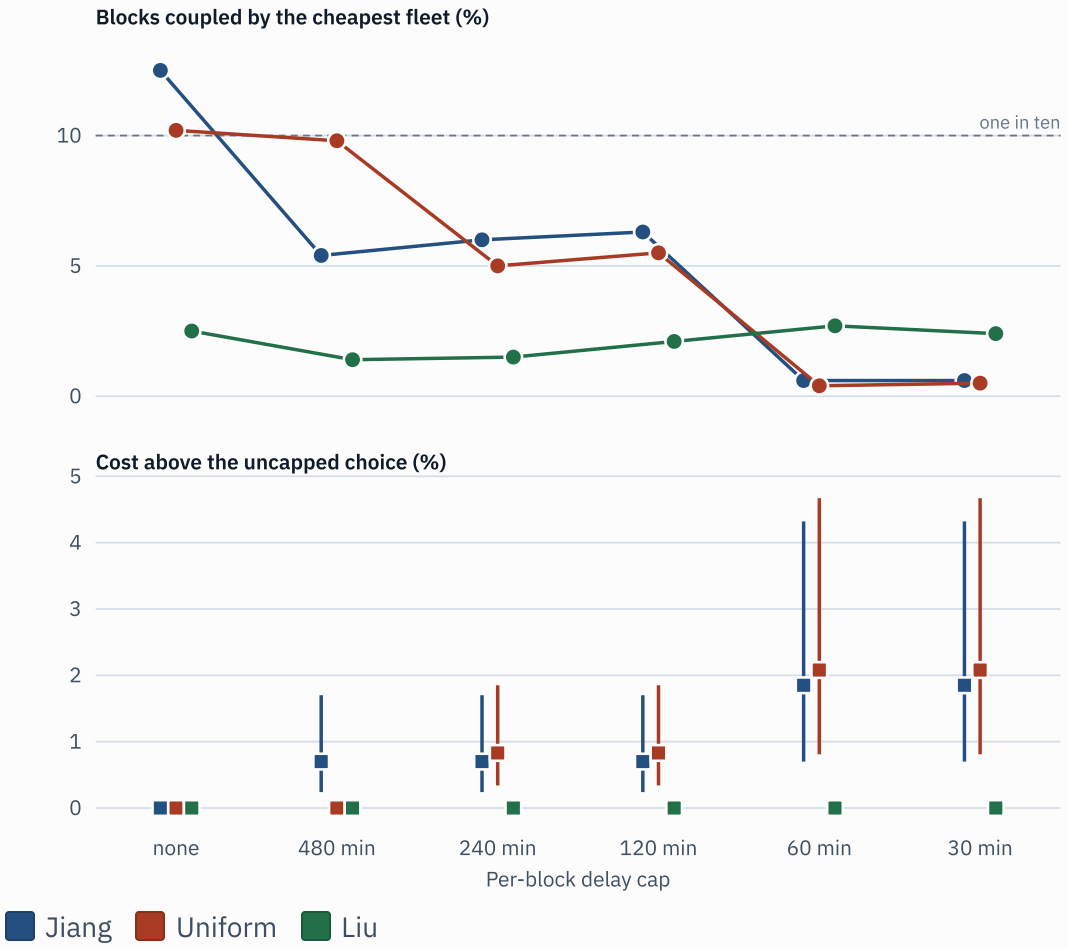
Most blocks more than 4 h late are coupled ones. Jiang masses, 300 t: 40 of 59. Jiang masses, 325 t: 16 of 23. Liu masses, 325 t: 24 of 26. In the mixes, the deferred blocks queue for the scarce heavy units.

With an average target alone, coupled lifts can slip indefinitely. No production plan can absorb that. So service needs a lateness cap. Tighter caps never lower a count.

Sources: this study, three mass profiles with short handling. Caps run from 30 min to none (paper §5.4, Fig. 10a).

A 120-min cap raises least cost by only 0–1.8%

Shift labour. Short handling, empirical due dates. 30 days per mass profile.



Masses	No cap	120-min cap	Cost
Jiang	5 × 300 t 12.5%	1 × 500 t + 4 × 270 t 6.3%	+0.2–1.7%
Uniform	2 × 500 t + 3 × 270 t 10.2%	2 × 500 t + 3 × 300 t 5.5%	+0.3–1.9%
Liu	2 × 425 t + 3 × 300 t 2.5%	2 × 425 t + 3 × 300 t 2.1%	0

- The cap excludes fleets that qualify by deferring. Under Jiang and uniform masses, this changes the purchase. The winner becomes a mix with heavy units that couples less.
- Without a cap, the winner couples ≤ 1/10 in only 75% of their decisions. With any cap from 480 down to 30 min, it does in all of them.
- Tightened to 30 min, the cap raises cost by at most 4.7%. Under Liu masses, 2 × 425 t + 3 × 300 t is cheapest at every cap.

Sources: this study (paper §5.4, Fig. 10b, c).

At a 90% or 98% target, the count barely moves

40 · 37

of 42 fleets keep their K* at a 90% · 98% target. The others move by one transporter.

93.3% • 87.1%

of 1,260 decisions keep the 95% winner. Outside the heavy tail, every winner couples ≤ 1/10 at both levels.

Why: the integer step of one transporter

- At K*, 36 of the 42 fleets already have schedules with at least 98% on time. A 37th falls one block short.
- One transporter fewer drops the pooled on-time share to 56–96%. 41 fleets can lose one transporter. For 32 of them, at least one day then has no schedule within the cap.

> **An average target cannot replace a limit on extreme lateness.**

Sources: this study, 42 fleets in seven conditions. They were re-sized at 90% and 98% on the original 30 days (paper §5.4).

Flexible coupling adds little, and only zero alignment lets a light fleet win

Two coupling rules

Flexible (main rule): any minimal team. **Restricted**: mixes couple only blocks no transporter of the fleet can carry alone. Restricted counts are pooled into the flexible results. Their schedules remain feasible.



Change of least cost under the restricted rule (shift labour, 1,620 decisions). At most 9.4%. Zero for the uniform, heavy-tail, Roh–Cha and Liu profiles. Largest for light-skewed (mean 1.4%). The mixes need flexible coupling: only 6 of 72 still qualify. Yet the decision hardly changes, because a single-carry fleet is nearly as cheap. Without pooling, search noise makes the restricted rule look cheaper in 8.6% of comparisons.

Alignment time

Jiang sample, short handling, empirical due dates, six fleet types. At δ = 0, a 300 t fleet coupling 12.5% wins 77 of 180 cost scenarios. From δ = 2.5 min it wins none. The winner becomes 270 t with one heavy unit (coupling 6.3–6.9%). From δ = 20 min it has two heavy units (≤ 0.2%).

88.6% → 100%

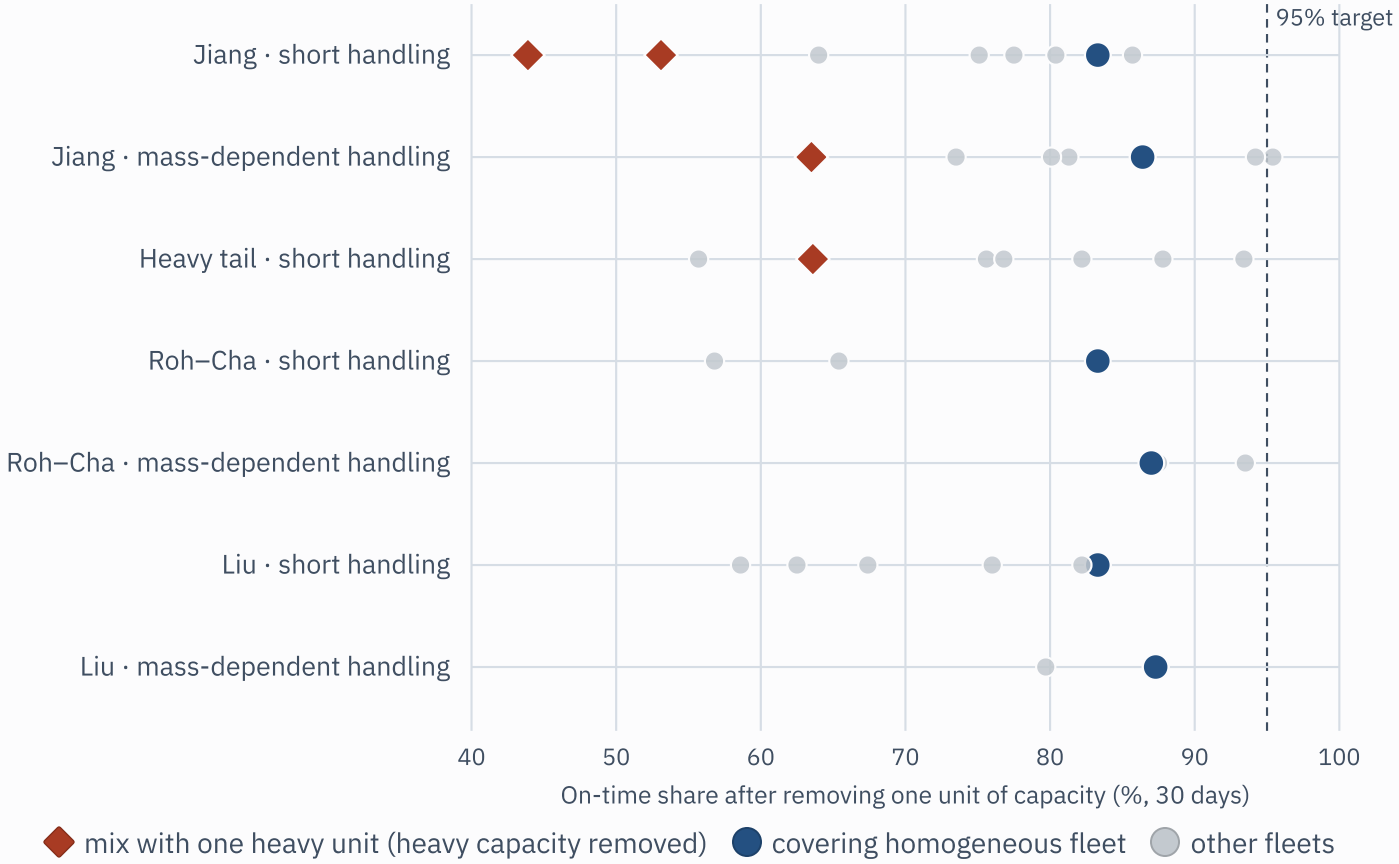
low-coupling winners in the one-factor runs (reduced candidate set). 88.6% at δ = 0. 100% at δ = 20–40 min.

> **Coupling pays only when it is almost free.**

Sources: this study (paper §5.4; Supplementary S2). The restricted rule follows Jiang et al. (2021). Alignment time has no published measurement. The tested range is 0–40 min.

Capacity-removal estimate: the cheapest redundancy is a second heavy unit or one more transporter

On-time share over 30 days after removing one unit of capacity. Each capacity class loses one unit; starts are reassigned. 42 contending fleets in seven conditions.



44–64% · 83–87%

outside the heavy tail. Single-heavy mixes without their heavy capacity · covering homogeneous fleets with one fewer unit.

- Redundancy is estimated from the qualifying counts of the original and reduced compositions. At those counts, we schedule normal operation. 87.5% of the 6,480 winners couple ≤ 1/10 (98.8% outside the heavy tail). Least cost is a median 8.8% above the nominal winner. Outside the heavy tail, no single-heavy mix wins. Such mixes were 5.8% of nominal winners.
- The option to couple is worth about as much as nominally. Its 90th-percentile upper bounds are 6.4–6.7%. Larger fleets lost relatively less service in only 3 of 7 conditions.

This is a capacity-composition estimate. Start positions are reassigned, not kept for the surviving vehicles. So it does not establish service after an arbitrary vehicle fails.

Sources: this study. Reduced compositions were rescheduled on the 30 original days. They use the start positions of their own scans. Redundancy counts come from the stored runs (paper §5.4; Supplementary S2, S5).

How reliable are the results? We looked for what could overturn them

① Other working days

Draw 30 days from the 30 **with replacement**. Compare all fleets again. Repeat 2,000 times (the bootstrap). Count how often the result holds.



Top: the original 30 days. Bottom: one redraw; yellow = days drawn more than once.

② Distance to a flip

Counts come from search. With more search, a loser might manage with one unit fewer and overtake. We count the on-time blocks it still lacks at one fewer. We compare that with what two extra searches recover.

strong: short by > 92 **medium: 59–92**

weak: ≤ 58

92 = the most on-time blocks two extra searches recovered in any fleet (30 days). 58.9 = its 95th percentile.

③ Disturb the inputs

Multiply every loading and unloading time by a random 0.5–1.5 (±50%). Use 20 scenarios a day. Check whether fleets sized to the target still suffice. Then re-size them to meet it under noise.

handling × U(0.5, 1.5) · 20 + 20 scenarios a day

We also checked exact counts on small instances and 30 fresh days. At the boundary, we ran six more searches a day. We compared counts from shallow search. The worst case gives every coupling fleet one unit fewer. Unmeasured parameters, manufacturer speeds, capacity removal and a second yard were tested too. All 214,210 scheduling runs are kept and can be recomputed.

Sources: bootstrap Efron & Tibshirani (1993). Grading thresholds and noise tests are ours (paper §3.4; Supplementary S2, S4). The ±50% noise level is assumed.

Checks of the findings: they change which fleet wins, not whether it carries alone

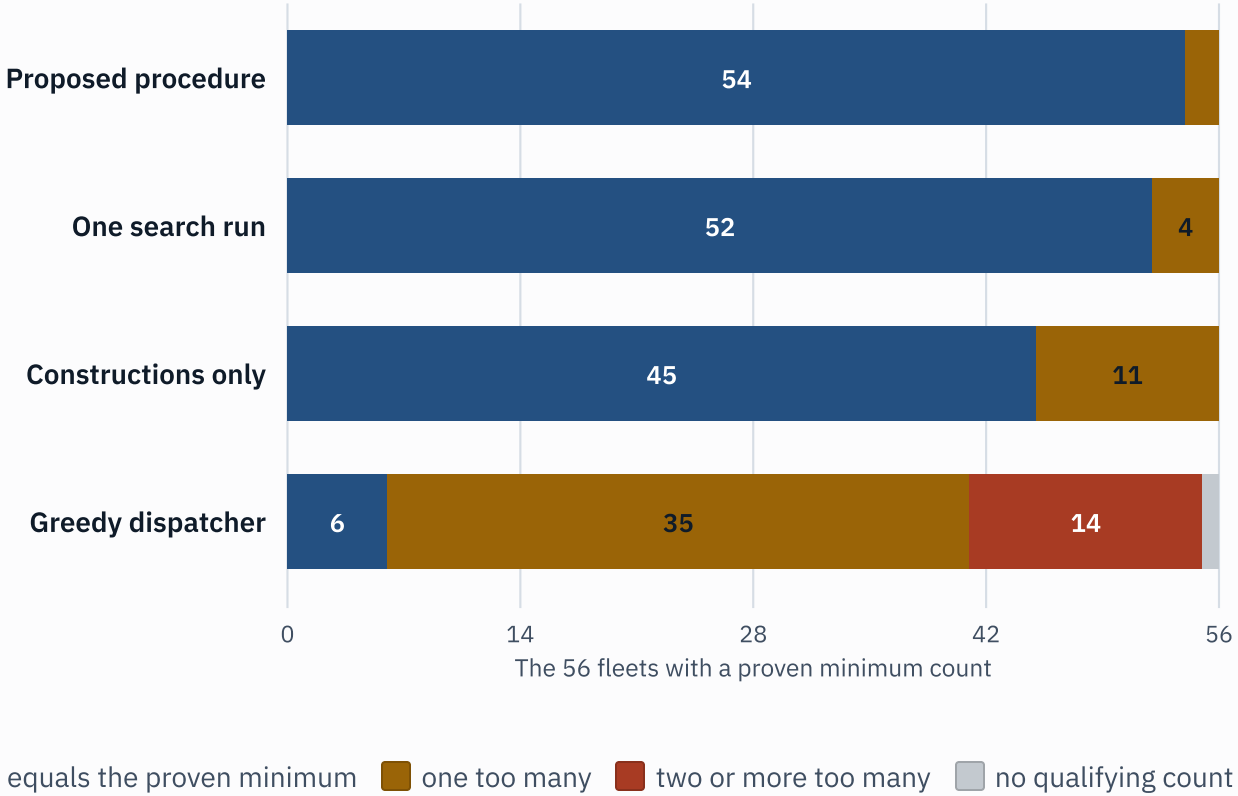
CHECK	WHAT CHANGES	LOW-COUPLING SHARE (OUTSIDE THE HEAVY TAIL)	OTHER OUTCOME
As reported, shift labour	—	91.0% (100%)	1,620 decisions
As reported, all labour measures	four labour measures	90.8% (99.4%)	6,480 decisions
Search depth	Counts after 0, 100 and 300 iterations. One search a day.	90.2 · 88.8 · 89.3 · 90.8%	same winner 77.5 · 81.1 · 89.4 · 95.0%
Boundary re-search	six more searches a day at K* – 1 for 66 fleets and 10 controls	90.8%	K* fell for 3 fleets (no control). 242 decisions (3.7%) changed.
Exact benchmark	80 single-batch slices. 70 fleets in 8 conditions.	—	Count equals the exact count for 54 of 56 proven fleets, one higher for 2. Certified winner in all 360 shift-labour decisions.
Adversarial bound	all 258 coupling-reliant fleets given one transporter fewer	68.8% (80.9%)	worst case
Day resampling	2,000 bootstrap samples of the 30 days, stored schedules re-used	89.2% (> 99.9%)	K* unchanged in 89.0% of resamples on average (median 97%)
Fresh days	30 new days. 42 fleets in 7 conditions.	98.6% (98.3%)	Same winner in 86.5% of 1,260 decisions. K* within one for all 42.
Slow speed and long handling	Adds 425 t, 500 t and mixes. Handling ×1.5 and ×2.	100% of 3,060 resolved	540 decisions outside the specification for every candidate
Capacity removal	reduced compositions, starts reassigned	87.5% (98.8%)	Same winner in 78.6%. Least cost +8.8% (median).
Second case	Liu et al. (2022) data, sized from scratch	100%	425 t cheapest in 179 of 180 decisions

Search dependence is deliberately strict. A decision counts if any loser is cheaper with one transporter fewer. That is 5,033 of 6,480 (77.7%). It is not a share of wrong decisions. Graded at K* – 1: 3,513 strong, 298 medium, 1,222 weak. All 1,447 certain decisions have a winner coupling ≤ 1/10. Among search-dependent ones, 88.1% do.

Sources: this study (paper §5.5, Table 6; Supplementary Table S1).

Exact benchmark: the procedure finds the proven minimum for 54 of 56 fleets

Each method's count against the proven minimum. Single-batch slices; 56 fleets have a proven minimum.



54 / 56

fleets have the proven minimum count. Two are one higher. None is lower.

360 / 360

shift-labour decisions pick a certified winner. All of them are close.

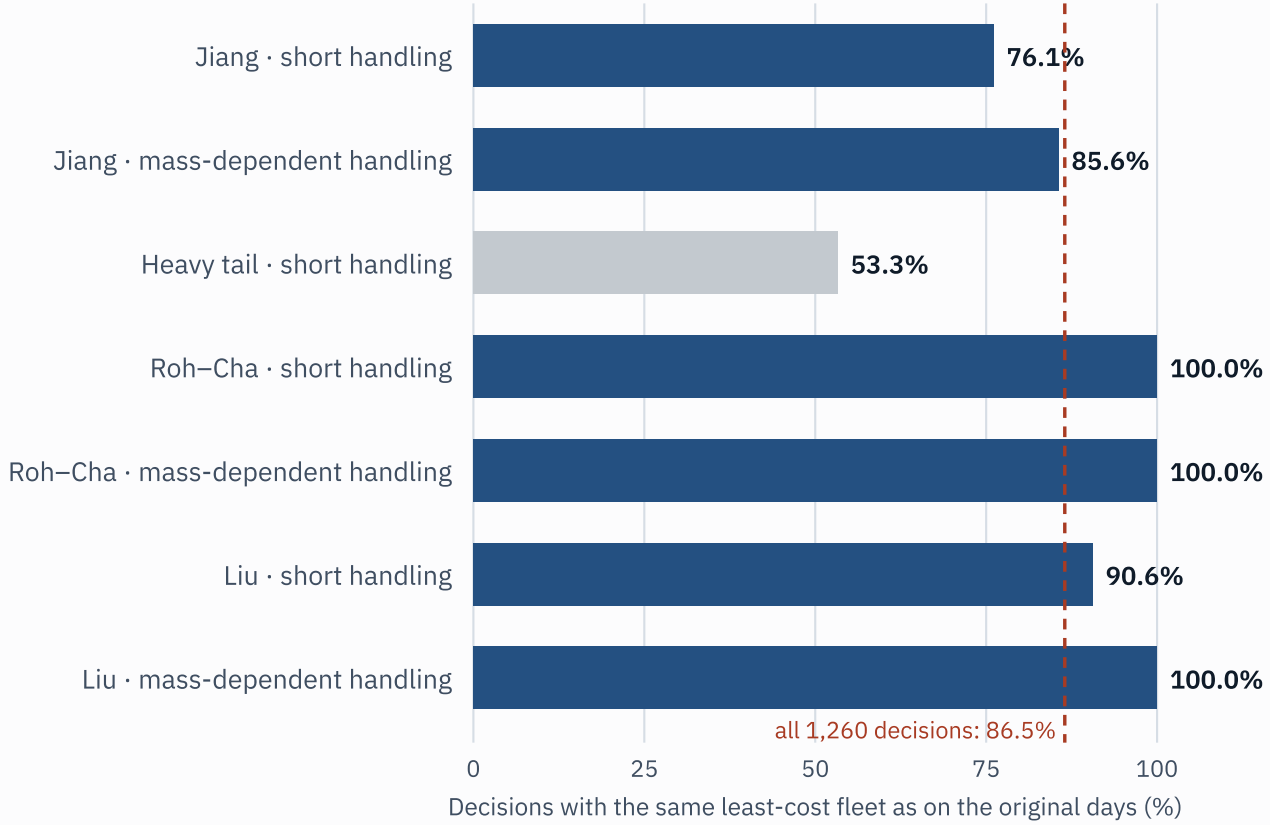
- There are 1,147 proven slice–count pairs. One search run finds the fewest late blocks in 94.1%. It is never more than one block above.
- Constructions alone give 11 of 56 counts too high. A greedy dispatcher gives 49, and one more has no count. They change the cheapest fleet in 37.5% and 25% of decisions. The extra cost reaches 4.5% and 11.9%.

> **Fleet design needs good scheduling. Cruder dispatching overstates the fleet.**

Sources: this study. 80 single-batch slices of 12–13 blocks, from ten days in eight conditions. CP-SAT of Google OR-Tools finds the fewest late blocks at each count. Paper §4.2, §5.5; Supplementary S7.

On 30 fresh days the kind of fleet holds

Share of decisions with the same least-cost fleet as on the original days. 180 decisions per condition.



39 / 42

fleets keep their K*. The rest move by one: two fell, one rose.

98.3%

of 1,080 decisions outside the heavy tail have a winner coupling ≤ 1/10. Over all 1,260: 98.6%.

- The named winner is the same in 86.5%. The largest change is in the heavy tail. There a 500 t fleet needed one transporter fewer. It wins 84 of 180 cost scenarios. Before, 550 t or 270 t with two heavy units had won them.
- With the original counts, 41 of 42 fleets still qualify. The exception is Roh–Cha 500 t with short handling. Six units have no capped schedule on one day. Seven qualify.

Sources: this study. The 30 days are not used anywhere else. Here 42 fleets contend for least cost in seven conditions. The service specification is the same. Paper §5.5; Supplementary S5.

Manufacturer speeds: 97% of winners still couple ≤ 1/10

Qualifying count K*: common speed → manufacturer speeds. Yellow: changed. Grey: not rerun, count kept.

Condition (empirical due dates)	380 t	425 t	500 t	550 t	270 t + 1 heavy	270 t + 2 heavy
Jiang · short	**5 → 6**	5 → 5	5 → 5	5 → 5	**5 → 6**	5 → 5
Jiang · mass-dep.	8 → 8	7 → 7	7 → 7	7 → 7	9 → 9	8 → 8
Roh–Cha · short	**8 → 9**	8*	6*	5 → 5	10*	**8 → 9**
Roh–Cha · mass-dep.	16 → 16	16*	12*	9 → 9	19*	**17 → 18**
Liu · short	**6 → 7**	5 → 5	5 → 5	5 → 5	**8 → 9**	**6 → 7**
Liu · mass-dep.	**10 → 11**	8 → 8	8 → 8	8 → 8	14*	**12 → 13**

The 380 and 425 t units run at 10/5 km/h. The 500 t, 550 t and mix units run at 12/5 km/h. A faster vehicle never makes a block later. So common speeds are the optimistic end. Mixes at the heavy speed are the pessimistic end.

45 / 270

shift-labour decisions change their cheapest fleet. All are Jiang masses with short handling. The one-heavy mix needs one more vehicle, and the two-heavy mix replaces it. Keeping the old choice costs up to 18.6% more.

178 / 1,080

decisions change their cheapest fleet at the pessimistic end (114 at the optimistic end)

1,054 / 1,080

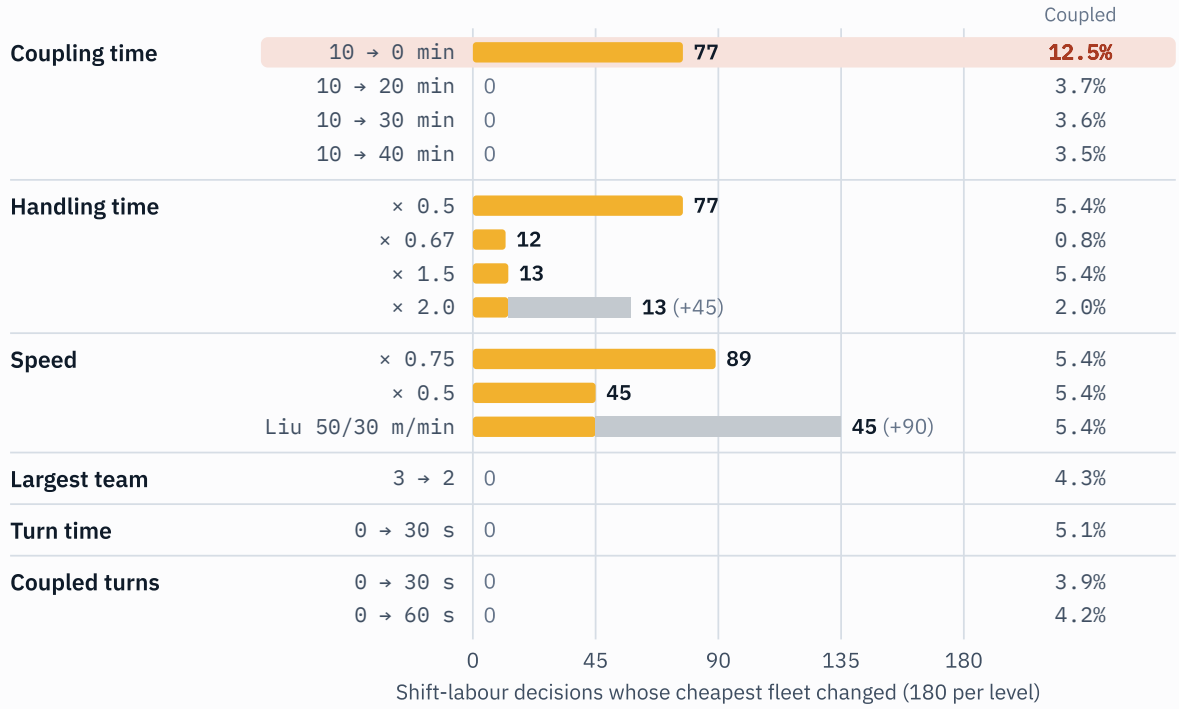
decisions where the cheapest fleet still couples at most one block in ten (pessimistic end)

- Under Roh–Cha masses, 550 t is cheapest in all 360 decisions. Under Liu masses, 425 t is cheapest in 350 of 360. The comparison covers the eleven reference fleets.

Sources: speeds from Suzhou Dafang data sheets. DCY270 runs at 12/6, DCY380 at 10/5 and DCY500 at 12/5 km/h. The 425 t and 550 t units take the nearest model (assumed). Results: this study (paper §5.6; Supplementary Table S1).

Unmeasured parameters, one at a time: choices move only to neighbours

Decisions whose cheapest fleet changed when one parameter changed. Affine price · shift labour. Right: the largest share the cheapest fleet coupled.



Grey: no fleet qualified within four units above its start, all with mass-dependent handling. We added 425 t, 500 t and mixes and scanned to 40 units. Every resolved winner then couples at most 6.2% (3,060 decisions). In the other 540, no candidate qualifies up to 40 units. They lie outside the conclusions.

98.3%

of 10,260 decided decisions: the cheapest fleet couples ≤ 1/10

12.5%

With zero alignment time, 300 t is cheapest and couples 12.5%. This is Jiang masses with short handling. From 2.5 min, a mix with heavy units wins.

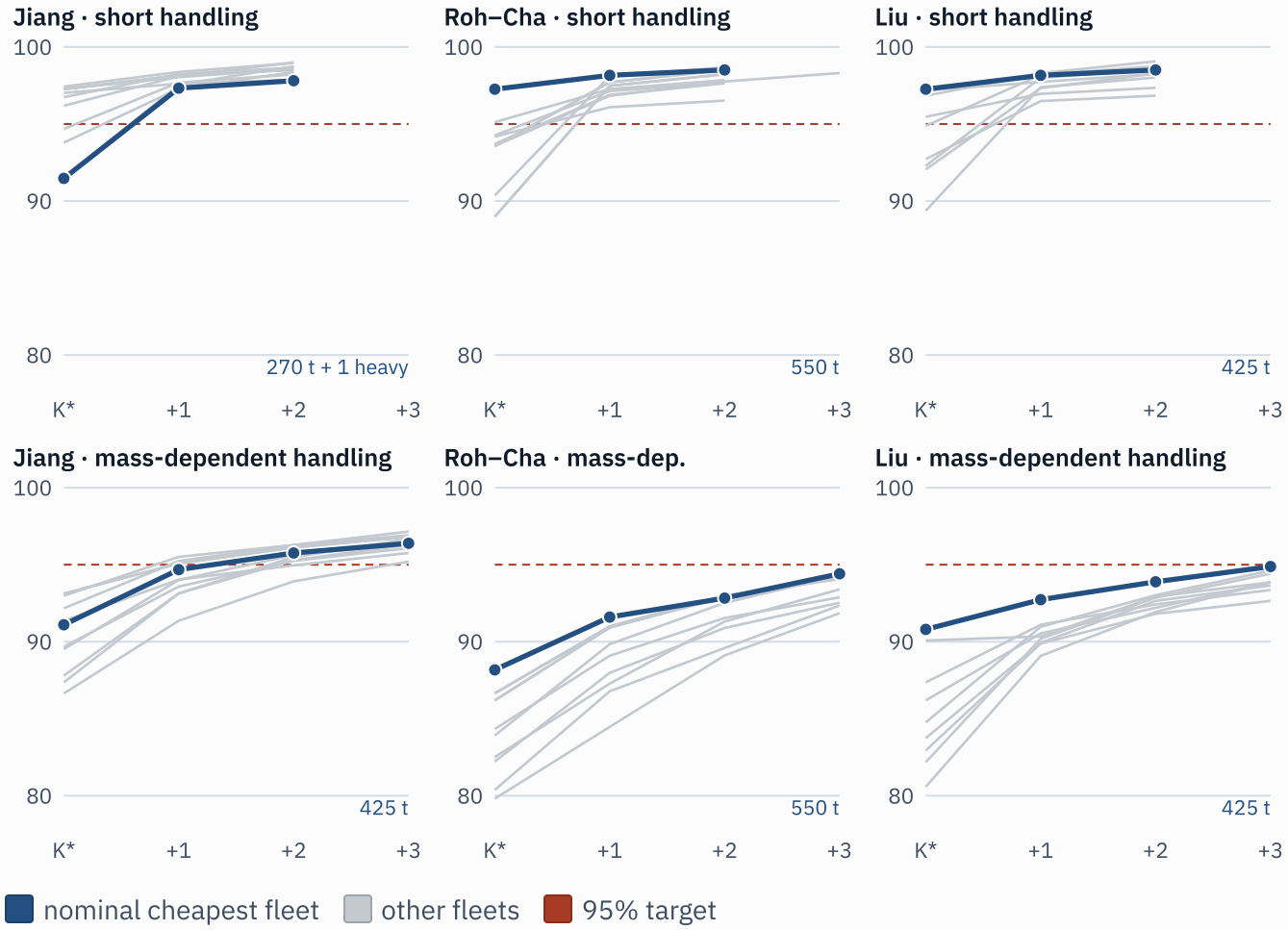
- Every change is under Jiang masses, between neighbouring fleets. Under Roh–Cha masses, 550 t never changes.
- With one crew per coupled team, 300 t also wins some decisions. This happens for Jiang with longer handling or slower travel.

Scope: 4 conditions (Jiang, Roh–Cha masses × short, mass-dependent handling). Five reference fleets: 270, 300, 380, 550 t and 270 t + 2 heavy. Alignment was also run at 2.5 and 5 min with six fleets.

Sources: tested levels on “Parameter sources (2)”. Liu's 50/30 m/min speeds: Liu et al. (2022) §4.1.1. Results: this study (paper §5.4, §5.5; Supplementary S3).

Under handling noise the winner changes but still carries alone

On-time share under ±50% handling noise, adding units from K*. One line per fleet. Schedules are chosen on 20 scenarios and tested on 20 others.



- Fleets sized exactly to the target are lean. At K*: 89.0–97.4% with short and 79.8–93.1% with mass-dependent handling. Only 15 of 66 fleets keep 95%.
- With short handling, one extra unit suffices. With mass-dependent handling, Jiang masses need one to three. None of the 22 Roh–Cha and Liu fleets gets there within three.

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decided decisions change their winner. For Jiang with short handling, the two-heavy mix replaces the one-heavy mix. With mass-dependent handling, 500 t replaces 425 t.

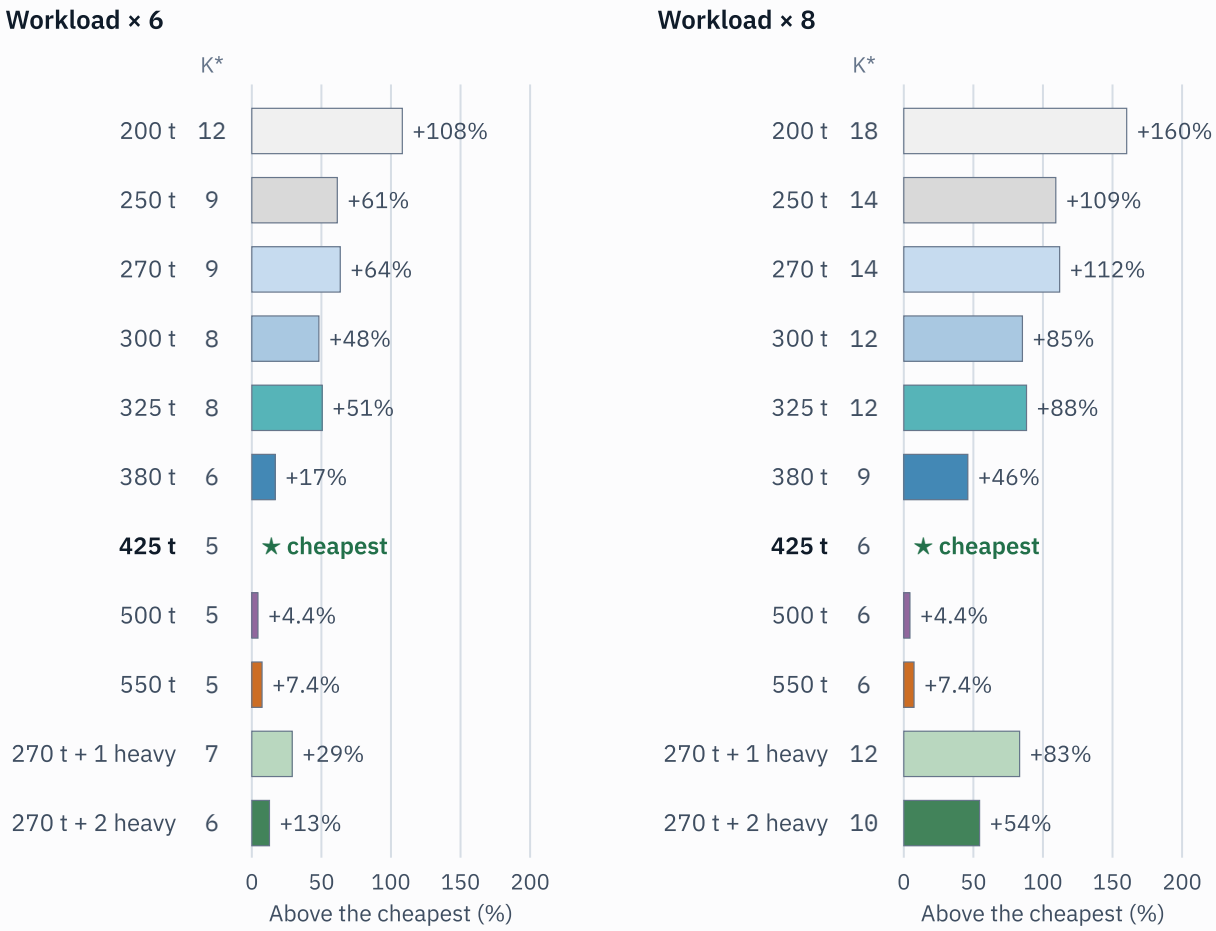
≈ 0 blocks coupled by the winners under noise (≤ 0.1%)

At the noise-robust counts, 42 of 44 fleets keep every block within the 120-min cap. The other two exceed it on 0.8% of scenario-days.

Sources: ±50% and 20 + 20 scenarios are our test design. Results: this study (paper §5.5; Supplementary Fig. S1, S4).

A second yard: 425 t is cheapest and couples nothing

Price ratio θ 12.4



The data come from a Shanghai cruise-ship yard. They include a week of transfers, a distance table and five units of 250–420 t. Demand is scaled by k = 1–8, about 10·k moves a day. There are 30 days and a 10-h shift. This is a reconstructed reference case, not a deployment.

- **Sized from scratch** (k = 6, 8): 425 t needs 5 and 6 units and couples nothing. A 270 t fleet needs 9 and 14. The 425 t fleet is cheapest in 179 of 180 decisions, by up to 10.6%.
- **Adding to the existing five**: they suffice up to twice the demand. From 4× without coupling, light additions never qualify. One or two 420 t units do. At 8× this needs after-shift work.

Mixes' heavy units are scheduled as 550 t and priced at 425 t. The remaining decision is at six times the demand. There 270 t with two heavy units is 0.5% cheaper. It couples 7.6% of blocks.

Sources: data from Liu et al. (2022), Tables 4–5 and supplementary data S1 (PLOS ONE, CC BY). The 10-h shift and demand multiples are assumptions. Decisions: five price curves × nine values of θ × shift labour and crew-vehicle hours. Results: this study (paper §5.5, Table 7; Supplementary S6).

When does coupling pay? What are the limits?

Four situations where coupling pays or is needed

- **No tier can carry the block**: coupling is necessary. This is the heavy-tail profile.
- **Near-instant alignment**: at δ = 0, a coupling 300 t fleet wins 43% of one condition's scenarios. At 2.5 min it wins none.
- **One crew drives a coupled team**: crew-team hours give the lowest low-coupling share (86.9%). With a covering tier, only they switch inside the calibrated range.
- **Capital far dearer than crews**: beyond about 1.4 crew-days per transporter-day. Remote or automated transporters could approach this.

> **Under the conditions studied, least-cost purchasing favours single-carrier capacity. Coupling is kept for exceptional blocks. Redundancy favours a second heavy unit or one more transporter.**

Limitations

- Costs are capital and labour only, without maintenance, energy or resale. Capacity loss uses reduced compositions with reassigned starts. Service after an arbitrary failure is unverified.
- Labour uses four stylised measures, not a crew roster. Reference mixes' heavy units are priced at the smallest tier they need, favouring mixes.
- Demand is stationary. Seasonal peaks, rental and staged expansion are out of scope.
- Paths are static shortest paths, without traffic or team conflicts. Joint control, load distribution, width, clearance, gradients and wheel loads are unmodelled. Width, clearance and wheel loads also limit the heaviest carriers; net effect untested.
- Process routes (shops, stockyard, dock) and road paths are inputs, not optimised. Joint optimisation with the fleet is future work.
- Whole-day K* are the smallest counts found, not certified minima.
- Excluded: slow-speed and long-handling cases where no candidate qualifies.

Sources: this study (paper §5.6, §6).

Conclusions: three answers and future work

(i) Least-cost fleets rely little on coupling

This holds under full-shift staffing. The winner couples at most one block in ten in 91.0% of 1,620 decisions. Outside the heavy tail, it does in all. In 13 further conditions, 300 t units with covering heavy units win 95.3% of 2,340 decisions. The handling-share rule sets their heavy units. All these winners couple at most one in ten.

(ii) Coupling competes only within a small budget of workload

Assume sub-additive prices and per-vehicle crews. Then a large-enough single carrier has the lower fluid cost per block. A light homogeneous fleet is cheaper only while its coupled share stays below the budget. Under full-shift staffing and the fitted power law, the budget is 2–18% of workload. Repeated service and alignment are a median 89.8% of light fleets' extra transporter-time.

(iii) A lateness cap affects both size and composition

Without a cap, fleets qualify by deferring their hardest blocks for hours. With the 120-min cap, six sized fleets need one more transporter. For the Jiang sample and the uniform profile, every winner then couples at most one block in ten. Without a cap, 75% do.

Scope

The conclusions rest on generated daily demand and static shortest paths. Coupled teams are assumed physically feasible. Whole-day counts are the smallest found. Exact single-batch benchmarks certify the least-cost fleet in all 360 shift-labour decisions. The cheapest tonnage is less stable than the kind of fleet. Close purchases should be reported as near-optimal sets. Reassess the coupling budget when capital-to-crew costs change.

Future work

- Check that coupled teams are physically feasible under road geometry and load distribution
- Optimise process routes together with the fleet
- Compare staged fleet expansion under seasonal demand

Sources: this study (paper §5.6, §6).

Status: all experiments are complete. The manuscript has been updated to version R03 and is being finalised.

Done

Main study: 36 conditions × 11 fleet types

Price and wage calibration

Search depth and boundary re-search

Exact benchmark (single batches)

Day resampling and 30 fresh days

Adversarial bound and evidence grading

Handling noise and robust counts

Lateness cap and on-time targets 90% / 98%

Heavy-block share and extended mixes (16 conditions)

Restricted coupling and alignment time

Manufacturer speeds, one-factor sensitivity, slow speeds scanned to 40

Capacity-removal estimate

Block-by-block decomposition

Second yard

Next

- Read the revised manuscript against the result tables
- Archive the released code and result tables with a DOI
- Submit to Ocean Engineering

Paper

Service-constrained fleet sizing for single-carrier and cooperative block transport in shipyards

214,210

scheduling runs, about 2,570 core-hours. All are reproducible.

Sources: this study's run records and the revised manuscript.

Five rules for planning a transporter fleet

1

Make every frequent block carriable by one transporter

The least-cost fleet is the smallest tier that carries (nearly) every block alone. Or it is light units with a few heavy ones. More capacity buys nothing. This qualifies the advice of Liu et al. (2022).

2

Use the coupling budget with calibrated occupancy

$h^* = \kappa \div (1 + 2\delta / \bar{w})$, about 2–18% under shift labour and $\alpha = 0.84$. Take occupancy from covering-fleet schedules. Exclude the lighter tier above a fluid cost ratio of 1.10.

3

Serve a thin heavy tail with a few heavy units, not coupling

Heavy units: $[K_{cov} \cdot h_H]$. This is within one unit in 96.6% of the decisions tested.

4

Compare spare capacity with coupling for redundancy

Blocks above every tier need coupling. In the capacity-removal estimate, least-cost redundancy is a second heavy unit or one more transporter. Failure protection must check each surviving vehicle's capacity and position.

5

Specify a lateness cap; report near-optimal sets

With an on-time share alone, fleets defer their hardest blocks for hours. Rankings are often within 1–2%. Report the near-optimal fleets and the decisions a better search could flip.

> **In one sentence: coupling is a remedy for exceptional blocks, not a purchasing strategy.**

Thank you. Questions and comments are welcome.

Sources: rules 1–5 rest on this study's results (paper §5.6). The advice of Liu et al. (2022) is in their discussion.

References and data sources (1/3)

1. Albert, B., Chaikovskaia, M., Gayon, J.-P., Quilliot, A. (2025). Sizing and scheduling for a fleet of reconfigurable mobile robots. *Int. J. Production Research* 63(18), 6758–6775. doi:10.1080/00207543.2025.2487565
2. Aziez, I., Côté, J.-F., Coelho, L.C. (2022). Fleet sizing and routing of healthcare automated guided vehicles. *Transportation Research Part E* 161, 102679. doi:10.1016/j.tre.2022.102679
3. Cao, Z., Zhang, S., Ceder, A. (2022). Novel coupling–decoupling strategy for scheduling autonomous public transport vehicles in overcrowded corridors. *Applied Mathematical Modelling* 106, 299–324. doi:10.1016/j.apm.2022.01.020
4. Chaikovskaia, M., Gayon, J.-P., Quilliot, A. (2026). Optimization of a fleet of reconfigurable robots. *Flexible Services and Manufacturing J.* 38(2), 470–492. doi:10.1007/s10696-025-09596-8
5. Drexl, M. (2012). Synchronization in vehicle routing. *Transportation Science* 46(3), 297–316. doi:10.1287/trsc.1110.0400
6. Efron, B., Tibshirani, R.J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.
7. Eskandari, H., Mahmoodi, E. (2016). A simulation-based multi-objective optimization study of the fleet sizing problem in the offshore industry. *Maritime Economics & Logistics* 18(4), 436–457. doi:10.1057/mel.2015.21
8. Fragapane, G., de Koster, R., Sgarbossa, F., Strandhagen, J.O. (2021). Planning and control of autonomous mobile robots for intralogistics: literature review and research agenda. *European J. Operational Research* 294(2), 405–426. doi:10.1016/j.ejor.2021.01.019
9. Gayon, J.-P., Lacomme, P., Oussama, A. (2025). Job-shop scheduling with cooperative transportation resources. *Int. Trans. in Operational Research*, Early View. doi:10.1111/itor.70145
10. Golden, B., Assad, A., Levy, L., Gheysens, F. (1984). The fleet size and mix vehicle routing problem. *Computers & Operations Research* 11(1), 49–66. doi:10.1016/0305-0548(84)90007-8
11. Guo, H., Wang, J., Sun, J., Mao, X. (2024). Multi-objective green vehicle scheduling in ship block transportation. *Scientific Reports* 14, 10796. doi:10.1038/s41598-024-61578-2
12. Hashemi Doulabi, H., Pesant, G., Rousseau, L.-M. (2020). Vehicle routing problems with synchronized visits and stochastic travel and service times: applications in healthcare. *Transportation Science* 54(4), 1053–1072. doi:10.1287/trsc.2019.0956
13. Hoff, A., Andersson, H., Christiansen, M., Hasle, G., Løkketangen, A. (2010). Fleet composition and routing. *Computers & Operations Research* 37(12), 2041–2061. doi:10.1016/j.cor.2010.03.015

Sources: DOIs checked against Crossref.

References and data sources (2/3)

14. Hu, Z.-H., Wei, C. (2018). Synchronizing vehicles for multi-vehicle and one-cargo transportation. *Computers & Industrial Engineering* 119, 36–49. doi:10.1016/j.cie.2018.03.023
15. Jiang, Z., Chen, Y., Li, X., Li, B. (2021). Multi-vehicle and one-cargo green transportation scheduling in shipbuilding. *Advanced Engineering Informatics* 49, 101306. doi:10.1016/j.aei.2021.101306
16. Joo, C.M., Kim, B.S. (2014). Block transportation scheduling under delivery restriction in shipyard. *Expert Systems with Applications* 41(6), 2851–2858. doi:10.1016/j.eswa.2013.10.020
17. Kim, B.S., Joo, C.M. (2012). Ant colony optimisation for block transportation scheduling with heterogeneous transporters. *Int. J. Production Research* 50(24), 7229–7241. doi:10.1080/00207543.2011.645078
18. Koç, Ç., Bektaş, T., Jabali, O., Laporte, G. (2016). Thirty years of heterogeneous vehicle routing. *European J. Operational Research* 249(1), 1–21. doi:10.1016/j.ejor.2015.07.020
19. Kweon, O., Kim, J., Kim, B., Kim, S.H. (2026). Pickup and delivery with time windows and two-dimensional loading for shipbuilding block transportation. *Computers & Operations Research* 194, 107573. doi:10.1016/j.cor.2026.107573
20. Liu, J., Yin, J., Khan, R.U. (2022). Scheduling management and optimization analysis of intermediate products transfer in a shipyard for cruise ships. *PLOS ONE* 17(3), e0265047. doi:10.1371/journal.pone.0265047
21. Nishi, T., Hiranaka, Y., Grossmann, I.E. (2011). A bilevel decomposition algorithm for simultaneous production scheduling and conflict-free routing for automated guided vehicles. *Computers & Operations Research* 38(5), 876–888. doi:10.1016/j.cor.2010.08.012
22. Pan, Z., Yu, Y. (2024). Optimizing heterogeneous multi-robot team composition for long-horizon construction tasks: time- and utilization-guided simulation. *Automation in Construction* 165, 105520. doi:10.1016/j.autcon.2024.105520
23. Park, C., Seo, J. (2012). A GRASP approach to transporter scheduling and routing at a shipyard. *Computers & Industrial Engineering* 63(2), 390–399. doi:10.1016/j.cie.2012.04.010
24. Pisinger, D., Ropke, S. (2007). A general heuristic for vehicle routing problems. *Computers & Operations Research* 34(8), 2403–2435. doi:10.1016/j.cor.2005.09.012
25. Roh, M.-I., Cha, J.-H. (2011). A block transportation scheduling system considering a minimisation of travel distance without loading of and interference between multiple transporters. *Int. J. Production Research* 49(11), 3231–3250. doi:10.1080/00207543.2010.484427

Sources: DOIs checked against Crossref.

References and data sources (3/3)

26. Ropke, S., Pisinger, D. (2006). An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation Science* 40(4), 455–472. doi:10.1287/trsc.1050.0135
27. Shin, J.-Y., Bak, N.-H. (2014). Transporter scheduling with transporter combination in shipbuilding. *J. Navig. Port Res.* 38(3), 299–305. doi:10.5394/kinpr.2014.38.3.299
28. Soares, R., Marques, A., Amorim, P., Parragh, S.N. (2024). Synchronisation in vehicle routing. *European J. Operational Research* 313(3), 817–840. doi:10.1016/j.ejor.2023.04.007
29. Vis, I.F.A. (2006). Survey of research in the design and control of automated guided vehicle systems. *European J. Operational Research* 170(3), 677–709. doi:10.1016/j.ejor.2004.09.020
30. Vis, I.F.A., de Koster, R., Savelsbergh, M.W.P. (2005). Minimum vehicle fleet size under time-window constraints at a container terminal. *Transportation Science* 39(2), 249–260. doi:10.1287/trsc.1030.0063
31. Voigt, S. (2025). A review and ranking of operators in adaptive large neighborhood search for vehicle routing problems. *European J. Operational Research* 322(2), 357–375. doi:10.1016/j.ejor.2024.05.033
32. Wang, C., Wang, K., Tao, J., Zhou, Y. (2020). Research on real-time optimal path planning model and algorithm for ship block transportation in shipyard. *J. Marine Science and Engineering* 8(12), 991. doi:10.3390/jmse8120991
33. Wang, N., Yin, J. (2026). Enhancing coastal industrial efficiency: a scheduling optimization framework for irregular assembly blocks in cruise ship manufacturing. *Frontiers in Marine Science* 13, 1843859. doi:10.3389/fmars.2026.1843859
34. Zhou, J., Wu, H., Weng, H., He, L., Li, W., Yang, T. (2025). Optimization of engineering vehicle scheduling in shipbuilding and repair yards based on the dual-cycle strategy. *Logistics* 9(4), 163. doi:10.3390/logistics9040163

Suzhou Dafang DCY270, DCY380, DCY500 data sheets (szdf.com, accessed 2026-09-24); TII Group transporter brochure 0317-EN; Google OR-Tools CP-SAT.

33 public award notices (csscbidding.com, China Shipbuilding Electronic Bidding Platform, 26; 365trade.com.cn, 5; e.sinochemitc.com, 2; 2022–2026), listed in Supplementary Table S4.

National Bureau of Statistics of China, 2025 average wages of urban employees (release of 15 May 2026), Table 7; China Statistical Yearbook 2025, Table 4-12; Shanghai HRSS (2025), State Council 2019 No. 13, MOHRSS 2015 No. 71, MOHURD 2016 No. 74.